Hyperspectral Anomaly Detection: A Dual Theory of Hyperspectral Target Detection

Chein-I. Chang, *Life Fellow, IEEE*

***Abstract*— Hyperspectral target detection (HTD) and hyperspectral anomaly detection (HAD) are designed by completely different functionalities in terms of how to carry out target detection. Specifically, HTD is a reconnaissance technique looking for known targets as opposed to HAD which is a surveillance technique seeking unknown targets of interest. So, HTD is generally designed by the hypothesis testing theory to derive likelihood ratio test (LRT)-based detectors. However, such hypothesis testing theory-based HTD requires the targets under the alternative hypothesis to be known. In addition, it also requires knowledge of the probability distribution under each hypothesis such as Gaussian distributions. Accordingly, the LRT-based HTD cannot be directly applied to HAD. This article develops a dual theory of LRT-based HTD for HAD, which converts HTD to HAD by making LRT-based detectors anomaly detectors. In addition, by virtue of this dual theory a new signal-to-noise ratio (SNR)-based theory can be also developed for HAD. Interestingly, the commonly used hyperspectral anomaly detector, referred to as Reed and Xiaoli detector (RXD), which is derived from the generalized LRT (GLRT), can be also rederived by this dual theory as well as the new developed SNR-based HAD theory.**

***Index Terms*— Anomaly detection (AD), generalized likelihood ratio test (GLRT), signal-to-noise ratio (SNR).**

NOMENCLATURE

2-D ROC	Two-dimensional receiver operating characteristic.
3-D ROC	Three-dimensional receiver operating characteristic.
ACE	Adaptive cosine estimate.
AMD	Adaptive matched detector.
AUC	Area under an ROC curve.
BKG	Background.
BS	Background suppression.
CEM	Constrained energy minimization.
DS	Data sphering/sphered.
DS-AD	SNR-derived anomaly detector in the data sphered-space.
GAD	Generalized anomaly detector.
GLRT	Generalized LRT.
HAD	Hyperspectral anomaly detection.
HTD	Hyperspectral target detection.
K/**K**	Covariance matrix.
K-AD	SNR-derived anomaly detector in the K-space.
K-SA	Spectral angle operated in K-space.
K-SA^2	Square of K-SA.
K-space	Data space is whitened by $\mathbf{K}^{-1/2}$.
LRT	Likelihood ratio test.
MLE	Maximum likelihood estimate.
NAMD	Normalized adaptive matched detector.
NMF	Normalized matched filter.
ODP	Overall detection probability.
OSP	Orthogonal subspace projection.
R/**R**	Correlation matrix.
R-AD	SNR-derived anomaly detector in the R-space.
R-space	Data space is whitened by $\mathbf{R}^{-1/2}$.
R-SA^2	Square of spectral angle operated in R-space.
ROC	Receiver operating characteristic.
RXD	Reed–Xiaoli's anomaly detector.
SA	Spectral angle.
SNPR	Signal-to-noise probability ratio.
SNR	Signal-to-noise ratio.
TD	Target detectability.
TDBS	Target detection and BKG suppression.
UHTD	Unsupervised HTD.
UITE	Unknown interesting target extraction.
UITF	Unknown interesting target finding.

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The author is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, and also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore, MD 21250 USA (e-mail: cchang@umbc.edu).

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I. INTRODUCTION

HTD is a fundamental task in hyperspectral data exploitation. According to [1, Ch. 5], HTD generally can be classified into two categories in the sense of detection modes, active target detection and passive target detection. Such taxonomy arises from a military perspective in many real world applications where active target detection is referred to as reconnaissance detection as opposed to passive target detection referred to as surveillance detection. In reconnaissance applications, the targets to be detected are always known, in which case, the target detection is carried out by actively detecting the desired targets specified by particular target knowledge provided by *a priori* information or visual inspection, referred to as *a priori* target detection, such as OSP [2]–[4], CEM subpixel target detection [3], [5], [6]. On many occasions,

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targets to be detected may not be available but can be obtained by unsupervised methods from the data to be processed. In this case, such obtained target knowledge can be considered as unsupervised target knowledge or *a posteriori* target knowledge. These unsupervised methods are then considered as UITE. The detection of this type is called *a posteriori* target detection, referred to as UHTD, such as unsupervised subpixel detection [3], [7], [8] and automatic target generation process (ATGP) [9]. So, both *a priori* target detection and *a posteriori* target detection perform HTD with the difference in how the target knowledge is obtained. Chang [10] developed three new theories from hypothesis testing, SNR, and SA perspectives which rederived HTD along with a comprehensive study and analysis on HTD.

In contrast to active target detection, passive target detection does not require any *a priori* or *a posteriori* target knowledge at all. It is particularly important in many surveillance applications with no known specific targets of interest, in which case target detection must be carried out passively in a completely unknown and blind environment. There are also two ways to perform passive target detection. One is generally known as UITF with no need of target knowledge, for example, *a posteriori* target knowledge obtained by UITE. Applications include endmember finding of rare material substances in geology, rescue and search in disaster areas, drug traffickers in law enforcement, and water/toxic waste pollution in Dianchi Lake, China, for environmental monitoring.

In addition to UITF, a second approach to passive target detection is HAD which does not find interesting unknown targets but rather finds targets with their characteristics distinct from their surrounding neighborhoods or environments. As a result, such found anomalous targets may not be all of interest such as BKG signatures. Applications include unexpected attack in a combat field, pesticide residual detection in food safety and inspection, soil contamination in agriculture, special species in ecology, tumor/mass detection in magnetic resonance imaging, and so on. A commonly used AD technique is the one developed by Reed and Xiaoli [12], known as RXD, denoted by $\delta^{\mathrm{RXD}}(\mathbf{r})$. There are two fundamental distinctions between UITF and HAD. One is that UITF finds one distinct target at a time, while HAD finds targets of all types at the same time including BKG signatures but cannot distinguish one type of targets from another. The other is that HAD may find many targets of the same type compared to UITF which generally finds one target for each different type as demonstrated in the hyperspectral digital imagery collection experiment (HYDICE) 15-panel experiments in Section X-A where HAD attempts to find all the panel pixels of different types instead of detecting individual different panel pixels. That is why HAD cannot be used for endmember finding. In this article, HAD is the main theme and was not investigated in [10].

There is also a key difference between UITE and UITF. Specifically, UITE assumes prior knowledge about the presence of targets, while UITF does not. As an example, it is a common misleading concept between endmember extraction and endmember finding in the literature. When endmember extraction is used, it has *a priori* assumption that there are endmembers present in the data to be processed so that an endmember extraction algorithm can be designed to extract endmembers. If there are no endmembers, what does an endmember extraction algorithm intend to extract? In this case, endmember finding fills in the gap since endmember finding does not assume the presence of endmembers *a priori* or *a posteriori*. It attempts to find material substances which are potentially considered as endmembers, many of which may be actually BKG signatures. This is why N-finder algorithm (N-FINDR) in [11] is called "N-finder" to find N endmembers as endmember finding but not "N-extractor" to extract N endmembers as endmember extraction. With this interpretation, ATGP in [9] performs actually endmember extraction as UITE, whereas the simplex growing algorithm (SGA) in [9] is considered as UITF in [1] and [9]. When the target presence is guaranteed, SGA becomes ATGP. Otherwise, ATGP becomes SGA. In other words, UITE makes use of the obtained *a posteriori* target knowledge to perform target detection as opposed to UITF which does not require the *a posteriori* target knowledge or any target model to perform target detection. The distinction also explains the difference between endmember finding and endmember extraction where the former does not assume the presence of endmembers as UITE, but the latter does as UITF. Unfortunately, in many works reported in the literature, endmember extraction is supposed to be endmember finding but has been widely misused and abused. More details can be found in [1].

As a real-world example in military applications, the concept of reconnaissance planes is fundamentally different from that of surveillance planes in terms of their missions. A U2 spy plane is considered as a reconnaissance plane because it has a specific mission corresponding to *a priori* target detection to carry out in active mode compared to an Airborne Warning And Control System (AWACS) plane, which is a mobile, long-range radar surveillance plane to fly over a region of interest to monitor and detect unusual military activities in passive mode without a specific mission which corresponds to AD. So, in terms of the terminologies suggested in this article, a U2 plane is used for UHTD as UITE and an AWACS plane is used for UITF.

Finally, we summarize several key differences between HTD and HAD as follows.

1) HTD assumes that the targets to be detected are known and provided by either *a priori* or *a posteriori* as described above, whereas HAD assumes no target knowledge at all. As a result, HTD can be formulated by a binary simple hypothesis testing with signals to be detected specified by the alternative hypothesis. By contrast, HAD does not have any knowledge about the targets that it intends to detect; it must be performed in a completely unsupervised manner. Consequently, the binary simple hypothesis testing is not applicable to HAD. To address this issue, it must utilize a binary composite hypothesis problem where the target to be detected is known by its target signature, but its signature intensity (abundance fraction) is unknown [13]. In this case, the unknown signature intensity will be estimated by a second reference dataset. The resulting detector is

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generally referred to as GLRT when the MLE is used to estimate the signature intensity for LRT resulting from a binary simple hypothesis-testing problem [14].

2) Because the target knowledge used by HTD is assumed to be known, the goal of developing a target detector should enhance TD as much as it can. Conversely, with unavailability of target knowledge HAD cannot enhance the targets in the same way that HTD does. Instead, it must rely on BKG /noise suppression to bring out the unknown interesting targets from BKG. So, in terms of searching targets of interest, HTD attempts to look for known targets. In doing so, the search strategy of HTD is to design a detection criterion which can find targets as effectively as possible. The desired detectors generally result in matched filters using various signals matched with the known targets. On the other hand, HAD does not search for specific targets because it does not know what targets it is seeking. In this case, HAD intends to suppress BKG as much as it can so that the unknown interesting targets can emerge from BKG as desired targets such as anomalies. Therefore, their primary tasks are different where HTD enhances TD as opposed to HAD which suppresses BKG to increase the contrast between anomalies and BKG.
3) As described above, HTD has specific targets to be detected. Its detection performance must be evaluated by its TD. On the other hand, HAD does not have specific targets of interest like HTD, but rather searches for targets with their spectral or spatial characteristics distinct from their surroundings. In this case, HAD must be evaluated by BKG suppressibility, not TD used for HTD. Interestingly, many works reported in the literature do not clearly clarify the differences between UHTD and HAD, and probably consider these two concepts are similar. Because of that, on many occasions UHTD and HAD have been analyzed and compared for performance analysis. As a matter of fact, such comparison is inappropriate like comparing apples to oranges. In particular, the detection measures used to evaluate the performance of HTD and HAD should be different with HTD evaluated by TD and HAD evaluated by the degree of BKG being suppressed. Accordingly, a direct use of the commonly used 2-D ROC for HTD may not be applicable to HAD.

In order to make the hypothesis testing-based LRT theory designed for HTD still work for HAD, the GLRT-based hypothesis testing has been also widely applied to HAD [12], [15]–[21]. Specifically, the adaptive matched filter (AMF) in [16] was used to derive $\delta^{\mathrm{RXD}}(\mathbf{r})$ in [12]. There are two issues arise from these approaches. One is the use of hypothesis testing theory which must assume a probability distribution under each hypothesis, generally Gaussian distributions. The other is the targets to be detected under the alternative hypothesis must either be known *a priori* or be estimated *a posteriori*. Unfortunately, anomalies cannot be obtained by either way. It seems that a direct use of the hypothesis testing theory is not feasible and also not realistic. Nevertheless, the GLRT approach developed in [12]–[13], [15]–[17], [20], and [22] still made an assumption that even through the target signal is not known, its target signature, $\mathbf{t}$ is known with its unknown signature abundance which can be estimated by the MLE through a second data set as a reference. Technically speaking, GLRT is a UHTD not HAD technique.

This article looks into a recent approach to HTD in [10] which models a subpixel target detection problem as a binary composite hypothesis-testing problem from a different aspect but similar to the one proposed in [12] and [13] that led to a GLRT-based anomaly detector $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$. In order to make an HTD-based subpixel active target detection problem an HAD-based subpixel passive target detection problem, a dual theory derived from HTD is developed for HAD, specifically, how to make an active HTD a passive HAD by converting the known target signature, $\mathbf{t}$ to an unknown target signature, $\mathbf{t}$. An interesting and novel idea is to consider a GLRT detector as a function of the data sample $\mathbf{r}$ and target signature, $\mathbf{t}$ as two dual variables, denoted by $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$. In HTD, we assume that the target signature $\mathbf{t}$ is given and known and then perform $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ on the data sample $\mathbf{r}$ which assumes to be a variable. However, in HAD, $\mathbf{t}$ is not supposed to be known in HAD, we perform $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ by simply interchanging the roles of $\mathbf{r}$ and $\mathbf{t}$ and operate $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ on $\mathbf{t}$ as a variable by assuming the currently being processed data sample $\mathbf{r}$ to be known and given. This dual-variable approach seems to be against the conventional wisdom in target detection, but it works. Specifically, if we define a GAD, $\delta^{\mathrm{GAD}}(\mathbf{r})$ by $\delta^{\mathrm{GAD}}(\mathbf{r}) = \max_{\mathbf{t}} \delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ with a given and fixed data sample $\mathbf{r}$, we can prove that $\delta^{\mathrm{RXD}}(\mathbf{r})$ is, indeed, a special case of $\delta^{\mathrm{GAD}}(\mathbf{r})$. To be more specific, RXD can be solved by finding the maximum of the dual-variable $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ over unknown target $\mathbf{t}$ by fixing the data sample $\mathbf{r}$. What is more interesting is that by taking advantage of the SNR theory developed in [10], we can even show that $\delta^{\mathrm{RXD}}(\mathbf{r})$ is actually $\max_{\mathbf{t}}\mathrm{SNR}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}))$. This fact paves the way that the SNR-based theory developed for HTD in [10] is readily applied to HAD. Most importantly, a close relationship between HTD and HAD via SNR is established and a dual theory of HTD in [10] can be also derived for HAD via the SNR-based HAD theory.

Unlike HTD which is primarily focused on TD, HAD should be evaluated by its ability in BS. Therefore, the traditional 2-D ROC curve is not appropriate to be used to measure the HAD performance. In this case, the 3-D ROC curve-based analysis developed in [23] can be used for this purpose where several detection criteria can be used to evaluate the effectiveness of BS.

Finally, it is worth noting that HAD presented in this article is completely different from the HTD in [10] in the sense that HTD is an active target detection with assuming that the targets to be detected must be known *a priori* as opposed to HAD which is a passive target detection without knowing the targets. As a result, their designing rationales are quite different. Interestingly, this article shows that HAD is in fact a dual theory of HTD developed by Chang [10] by interpreting the data sample $\mathbf{r}$ and target sample $\mathbf{t}$ as dual variables. In this case, we can make an active LRT-based target detector a passive target detector by interchanging roles of the data sample $\mathbf{r}$ and target sample $\mathbf{t}$ with $\mathbf{t}$ assumed to be unknown

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target to be detected for any given data sample $\mathbf{r}$. The resulting passive target detector turns out to be an anomaly detector. This explains why GLRT can be applied to HAD and leads to RXD.

As a conclusion, four main results of this article are summarized as follows:

1) We rederive RXD in Theorems 1 and 2 from a subpixel target detection problem via a binary composite hypothesis testing formulation [10]. Many researchers working in HAD community may have taken RXD derived from GLRT for granted [12], [13], [15]–[21]. This article offers a simple and intuitive approach to deriving GAD which helps to understand how RXD is derived from GLRT from a perspective of HAD.
2) A close relationship between GRLT and SNR is established in Theorem 3 to further show that GAD can be derived from maximizing the SNR of GLRT over all possible unknown target signatures $\mathbf{t}$ for every given data sample $\mathbf{r}$ without making Gaussian distributions. In other words, GAD is the maximum SNR version of GLRT.
3) A new SNR-based HAD theory is developed to rederive GLRT and GAD. Specifically, it includes GLRT as it is a special case.
4) A dual theory is derived in Theorem 4 to further explore the relationship between HTD and HAD.

The remainder of this article is organized as follows. Section II briefly reviews related works, describes motivations, and summarizes novelties of this article. Section III represents GLRT as a function of two parameters as dual variables, data sample, and target signature, and further proves that GLRT can be derived as an anomaly detector by Theorem 1 and Theorem 2. Section IV proves Theorem 3 to establish the relationship between GRLT and SNR. Section V develops an SNR theory for HAD. Section VI explores the connection between LRT-based detectors and SNR-based anomaly detectors. Section VII presents HAD as a dual theory of HTD via proving Theorem 4. Section VIII discusses the images to be used for experiments. Section IX conducts experiments and discusses results. Section X recaps and summarizes the work of this article.

II. Related Works, Motivations, and Novelties

HAD has been studied extensively in the past. The RXD developed in [12] is the most widely used anomaly detector and most currently being used anomaly detectors in the literature are either the variants of RXD or derived from RXD. So, only RXD-related works will be reviewed in this section.

A. Related Works

In many practical applications, *a priori* information about the targets of interest is usually not available such as anomalies or only partially available such as subpixel or mixed pixel targets. Under such circumstances, it is highly desirable to design a detector that can find targets whose spectral signatures are distinct from the spectra of their surrounding BKG pixels. A classic approach is to utilize statistical models to describe the BKG so that a signal detection model can be still used. In this case, a binary simple hypothesis testing is extended to a binary hypothesis testing formulation where the alternative hypothesis, H_1 and the null hypothesis, H_0 are used to characterize a known signal spectrum with unknown signal intensity and the BKG in which the target is embedded, respectively. With this formulation, a multivariate Gaussian distribution model is generally used to lead to RXD in [12] which has been widely used for AD.

An anomaly is generally referred to as a data sample whose spectral characteristics are distinctive from its surrounding data samples. How much distinction is sufficient to qualify a data sample as an anomaly is determined by a detector. A classic theory to derive HAD is to use the binary composite hypothesis testing theory where the null hypothesis, H_0, and the alternative hypothesis, H_1, represent BKG plus noise and a signal plus additive noise with each hypothesis governed by probability distributions, $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$, respectively. According to [12], the sought optimal detector is GLRT.

In the early development of RXD-based anomaly detectors [12], [13], [15]–[17], [20], [22], anomalies were considered as targets compared to their surrounding BKG which was modeled by structured or unstructured BKG. These BKG models were characterized by the reference data obtained from a local or global neighborhood of the data sample which can be described by Gaussian distributions. In particular, Stein *et al*. [13] assumed that BKG models could be characterized by three clutter models, the local normal model, the global normal mixture model, and the global linear mixture model from which GLRT could be derived for HAD. Specifically, anomalies were considered as unknown signals in a known target subspace. As the number of reference data samples grows indefinitely, the distributions can be approximated asymptotically by a Gaussian distribution. As a result, when a signal to be detected is specified by an unknown signal intensity parameter, the binary simple hypothesis-testing problem is generalized to a binary composite hypothesis-testing problem [14] with LRT being also extended to GLRT. Accordingly, GLRT has been the base for designing the AD algorithms over the past years in HAD community. For more details about RXD we refer to [12], [13], and [15]–[22].

Since RXD was developed from GLRT for HAD, many of its variants have been reported to address the above three models, local, global, and mixture models discussed above. For example, window-based local anomaly detectors implemented $\mathbf{K}$ or $\mathbf{R}$ using local windows [24], [25], sliding windows [26], dual windows [27], [28], and multiple windows [29]. In addition, kernel-based anomaly detector [30], AD for unlabeled classification [31], real-time processing of anomaly detection [32], guided filtering-based HAD [33], and spectral-spatial feature extraction-based HAD [34] were also developed.

However, as noted above, how to characterize BKG is the key element in designing HAD algorithms. In recent years, many approaches have been proposed along this line by either estimating or reconstructing BKG so that anomalies can be extracted after BKG being suppressed or BKG being removed, for example, BKG estimation [35]–[39], BKG estimation by Gaussian mixture model (GMM) [40],

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BKG separation [41], BKG estimation autoencoder [42], [43], adversarial autoencoder (AAE) in [44] and [45], generative adversarial network (GAN) [46]–[48], low rank and sparse representation [49]–[55], BKG-annihilated OSP [56]–[58], and deep learning convolutional neural networks [59]–[61].

B. Motivations

The motivations arise from how to address a signal detection in noise via a hypothesis-testing problem without knowing signals to be detected such as anomalies. In a binary hypothesis-testing problem, it generally assumes that the signal to be detected must be known and the probability distribution under each hypothesis must be also specified. In order to make the hypothesis testing theory applicable to an AD problem, these two issues need to be resolved. As reviewed in Section II-A, a general and classic approach assumes that an unknown signal is characterized by a known shape (signature) parameterized by an unknown signal intensity. That is, a hyperspectral target signal to be detected is specified by a known spectral signature with an unknown target abundance fraction. In addition, probability distributions under hypotheses must be also modeled by Gaussian distributions. In practical applications in HAD, neither assumption is true. First of all, the fact that a target is called an anomaly is generally because it cannot be identified by prior knowledge or visual inspection. Second, its existence is usually unexpected. Third, the probability of its occurrence is considerably very low. Most importantly, once it is present, its population is very small and cannot be described by a Gaussian distribution. Accordingly, from a perspective of HAD, the above GLRT approach is not appropriate and does not really solve the problem. However, does it imply that GLRT will not work for HAD? Obviously, this may be true from a perspective of target knowledge assumed to be known. So, we need to reinvent the wheel by coming up an idea of how to make an active LRT-based target detector a passive anomaly detector. To address this issue, a dual relationship between HTD and HAD should be explored and investigated.

C. Novelties

The following novelties are originated from addressing the two issues identified by motivations in Section II-B, none of which can be found in [10].

1) The most important and significant novelty is that GLRT will be shown to still work for HAD without assuming the target signature $\mathbf{t}$ to be known. This idea is well illustrated by the following example. It is similar to the extension of a binary Huffman code to a p-ary Huffman code with $p > 2$ information theory [62]. In order to avoid an unused codeword being occurred at an internal node of an incomplete p-ary Huffman coding tree, an introduction of a dummy variable is necessary to guarantee that the source alphabet with the least probability is always encoded at a leaf, not an internal node. This example provides an idea of how to use the unknown target signature $\mathbf{t}$ in GLRT as a hypothetical variable to be a dummy variable. Since a binary hypothesis-testing problem requires a known target signature $\mathbf{t}$ specified in the alternative hypothesis H_1 to be detected as an anomaly which is unfortunately not be known. To resolve this dilemma, GLRT is represented as a function of two dual variables, data sample $\mathbf{r}$ and target signature $\mathbf{t}$, denoted by $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$. Then, GLRT can be considered as a function of a hypothetical target signature $\mathbf{t}$ to be detected as a dual variable for any given fixed data sample $\mathbf{r}$. So, the dual variable$\mathbf{t}$ in $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ can be assumed to be a hypothetical dummy variable which will be eventually vanished and replaced by its optimal value, that is the maximum SNR of $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ over unknown target signature $\mathbf{t}$ for a given observed data sample $\mathbf{r}$ as proved in Theorem 1. It then continues to prove Theorem 2 that the maximizing $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ over $\mathbf{t}$ turns out to be GAD which does not depend on the target signature $\mathbf{t}$ which it intends to detect. Accordingly, RXD can be further shown to be a special case of GAD. As a matter of fact, this idea was also implicitly shared in deriving from GLRT to RXD [22] where it assumed that $\mathbf{S}_{\mathbf{t}}$ in GLRT [22, Eq. (19)] obtained from a reference data was a full-rank target signature matrix used to specify anomalies and was then canceled out to derive RXD [22, Eq. (20)].
2) Two proofs of Theorem 1 in Section III provide great benefits to those working in HAD on how to explore insights into the two well-studied concepts of data whitening and OSP in hyperspectral imaging.
3) Theorem 3 in Section IV further proves that GAD also maximizes the SNR of GLRT over all possible target signatures $\mathbf{t}$ for every given data sample $\mathbf{r}$ without making Gaussian distributions on $\mathbf{r}$. In other words, GAD is the maximum SNR version of GLRT over the dual variable, the unknown target signature $\mathbf{t}$.
4) Having Theorem 3 proved in Section IV paves a way for the development of SNR-based HAD to derive GLRT and GAD. Specifically, it includes GLRT as it is a special case. Specifically, it is shown in Section VII that GLRT can be applicable to HAD which leads to RXD.
5) Section VII proves Theorem 4 and also derives a dual theory of HTD for HAD by exploring a dual relationship between HTD and HAD. Specifically, we make an active HTD a passive HAD by interchanging the two dual variables, the unknown target signature $\mathbf{t}$ to be detected and the data sample $\mathbf{r}$ currently being processed. In this case, the traditional LRT-based target detectors can be still derived for AD.
6) The most crucial and substantial difference between HTD and HAD is how to measure their detection performance. Such a difference is similar to that between brightness and contrast in image processing. Since HTD is an active target detection which assumed the targets to be detected must be known, its performance should be evaluated by TD corresponding to target brightness. On the other hand, since HAD is a passive target detection and does not have any knowledge about the targets it attempts to detect, its effectiveness must be relied on BKG suppressibility (BS) corresponding to contrast

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resulting from BS not TD. To address this key issue, 3-D ROC curves are used to particularly design various TD-based and BS-based detection measures to evaluate and compare effectiveness of an anomaly detector in the performance of TD and BS. Specifically, it shows that the commonly used 2-D ROC curve of detection probability ($P_{\rm D}$) versus false alarm probability ($P_{\rm F}$) for AD performance is unreliable and may be sometimes misleading.

III. Generalized Likelihood Ratio Test Approach to Anomaly Detection

Since the original derivations of RXD involve a very complicated mathematical theory in statistical signal processing [12], [13], [15]–[21], it may present difficulty for those who are working in HAD community without decent BKG knowledge. Accordingly, many researchers have taken RXD for granted and very few intend to fully understand how RXD is derived. Specifically, how the target signature $\mathbf{t}$ with unknown signal intensity used in GLRT being taken care of.

This section presents a binary composite hypothesis testing theory that is different from the works in [12], [13], [15], [19], and [21] to derive the RXD. It formulates an AD problem as a hyperspectral subpixel detection problem via a binary composite hypothesis testing theory as follows.

Assume that $\mathbf{t}$ is an unknown target signature and the parameter α is an abundance fraction associated with $\mathbf{t}$ needed to be detected. A binary composite hypothesis-testing problem is cast by

$$\begin{aligned} &H_0 : \mathbf{r} = \mathbf{b} \\ &\text{versus} \\ &H_1 : \mathbf{r} = \alpha\mathbf{t} + \mathbf{b} \end{aligned} \tag{1}$$

where $\mathbf{r}$ is an L-dimensional hyperspectral pixel vector with L being the number of spectral bands and $\mathbf{b}$ is the BKG signature vector with its probability distribution described by a Gaussian distribution with the mean $\boldsymbol{\mu}_{\mathbf{b}}$ and covariance matrix $\mathbf{K}$. In this case, $p_0(\mathbf{r}) \sim N(\boldsymbol{\mu}_{\mathbf{b}}, \mathbf{K})$ and $p_\alpha(\mathbf{r}) \sim N(\alpha\mathbf{t} + \boldsymbol{\mu}_{\mathbf{b}}, \mathbf{K})$ are two probability distributions under H_0 and H_1, respectively. Due to the unknown parameter α in (1), the solution to (1) is GLRT, $\Lambda(\mathbf{r})$ given by

$$\begin{aligned} &\Lambda(\mathbf{r}) \\ &= \frac{\max_\alpha p_\alpha(\mathbf{r})}{p_0(\mathbf{r})} \\ &= \frac{\exp\left[-(\mathbf{r} - \hat{\alpha}_{\rm MLE}(\mathbf{r})\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r} - \hat{\alpha}_{\rm MLE}(\mathbf{r})\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})\right]}{\exp\left[-(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right]} \\ &\Leftrightarrow \log\Lambda(\mathbf{r}) = 2\hat{\alpha}_{\rm MLE}(\mathbf{r})\mathbf{t}\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) \end{aligned} \tag{2}$$

and

$$\delta^{\rm GLRT}(\mathbf{r}) = \begin{cases} 1, & \text{if } \Lambda(\mathbf{r}) \geq \tau \\ 0, & \text{if } \Lambda(\mathbf{r}) < \tau \end{cases} \tag{3}$$

that is,

$$\delta^{\rm GLRT}(\mathbf{r}) = \begin{cases} 1; & \text{if } \hat{\alpha}_{\rm MLE}(\mathbf{r})\mathbf{t}\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) \geq \tau' \\ 0; & \text{if } \hat{\alpha}_{\rm MLE}(\mathbf{r})\mathbf{t}\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) < \tau' \end{cases} \tag{4}$$

with $\tau' = (1/2)\log\tau$.

Since

$$\begin{aligned} \hat{\alpha}_{\rm MLE}(\mathbf{r}) &= \arg\left\{\max_\alpha p_\alpha(\mathbf{r})\right\} \\ &= \arg\left\{\min_\alpha (\mathbf{r} - \alpha\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r} - \alpha\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})\right\} \end{aligned} \tag{5}$$

the solution to (5) can be found as follows.

Differentiating $(\mathbf{r} - \alpha\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r} - \alpha\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})$ in (5) with respect to α and letting it be zero, that is,

$$\begin{aligned} &\left.\frac{\partial\left[(\mathbf{r} - \alpha\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r} - \alpha\mathbf{t} - \boldsymbol{\mu}_{\mathbf{b}})\right]}{\partial\alpha}\right|_{\hat{\alpha}_{\rm MLE}(\mathbf{r})} = 0 \\ &\Rightarrow -2\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) + 2\hat{\alpha}_{\rm MLE}(\mathbf{r})\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t} = 0 \end{aligned}$$

results in

$$\begin{aligned} &\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) = \hat{\alpha}_{\rm MLE}(\mathbf{r})\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t} \\ &\Rightarrow \hat{\alpha}_{\rm MLE}(\mathbf{r}) = \frac{\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}} = \frac{(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}\mathbf{t}}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}}. \end{aligned} \tag{6}$$

Substituting (6) in (4) yields

$$\delta^{\rm GLRT}(\mathbf{r}) = \begin{cases} 1; & \text{if } \dfrac{\left(\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right)^2}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}} \geq \tau' \\ 0; & \text{if } \dfrac{\left(\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right)^2}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}} < \tau' \end{cases} \tag{7}$$

which can be considered as an unsupervised LRT-based target detector for UHTD with its detector statistics obtained by $\hat{\alpha}_{\rm MLE}(\mathbf{r})$ in (6).

The GLRT in (7) is also called AMF in [16] along with its square root also being called matched filter (MF) [18].

Since the $\delta^{\rm GLRT}(\mathbf{r})$ in (7) is actually a function of two variables, data sample $\mathbf{r}$ and target signature $\mathbf{t}$, it is more accurate to re-express $\delta^{\rm GLRT}(\mathbf{r})$ as $\delta^{\rm GLRT}_{\mathbf{t}}(\mathbf{r})$ with $\mathbf{r}$ and $\mathbf{t}$ as dual variables. In this case, we can prove the following theorem by representing $\delta^{\rm GLRT}_{\mathbf{t}}(\mathbf{r})$ in two different forms by two different techniques—data whitening and OSP.

Theorem 1: For two dual variables $\mathbf{r}$ and $\mathbf{t}$

$$\text{(a)}\quad \delta^{\rm GLRT}_{\mathbf{t}}(\mathbf{r}) = \frac{\left\|\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right\|^2}{||\mathbf{K}^{-1/2}\mathbf{t}||^2} \tag{8}$$

$$\text{(b)}\quad \delta^{\rm GLRT}_{\mathbf{t}}(\mathbf{r}) = \left\|P_{\mathbf{K}^{-1/2}\mathbf{t}}\left(\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right)\right\|^2. \tag{9}$$

Proof of (a): Now we rewrite GLRT in (7) as follows. Since $\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})$ is a scalar, $(\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}))^T = \mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})$, which implies that $\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})$ is the same as $\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\mathbf{t}$. So, the detector statistics in (7) can be re-expressed as

$$\begin{aligned} \delta^{\rm GLRT}_{\mathbf{t}}(\mathbf{r}) &= \frac{\left(\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right)^T\left(\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right)}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}} \\ &= \frac{(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}\mathbf{t}\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}}. \end{aligned} \tag{10}$$

The numerator of (10), $(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}\mathbf{t}\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})$, can be re-expressed as

$$\begin{aligned} &(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}\mathbf{t}\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) \\ &= (\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1/2}\left(\mathbf{K}^{-1/2}\mathbf{t}\right)\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) \\ &= \left[\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right]^T\left[\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right] \\ &= \left\|\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right\|^2. \end{aligned} \tag{11}$$

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It should be noted that $\mathbf{K}^{-1/2}\mathbf{t}$ in (11) can be considered as the whitened $\mathbf{t}$ via the whitening matrix, $\mathbf{K}^{-1/2}$.

The denominator of (10), $\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}$ can be re-expressed as

$$\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t} = \left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\left(\mathbf{K}^{-1/2}\mathbf{t}\right) = \left\|\mathbf{K}^{-1/2}\mathbf{t}\right\|^2. \quad (12)$$

Substituting (11) and (12) into (10) yields (8).

Proof (b): By sandwiching the scalar $(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t})^{-1}$ between $\mathbf{t}$ and $\mathbf{t}^T$ in (12) we can obtain $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ in an alternative form

$$\begin{aligned}\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}) &= (\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}\mathbf{t}\left(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}\right)^{-1}\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \\ &= \left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right)^T\left(\mathbf{K}^{-1/2}\mathbf{t}\right)\left(\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\left(\mathbf{K}^{-1/2}\mathbf{t}\right)\right)^{-1}\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T \\ &\quad\times\left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right). \quad (13)\end{aligned}$$

Define

$$P_{\mathbf{K}^{-1/2}\mathbf{t}} = \left(\mathbf{K}^{-1/2}\mathbf{t}\right)\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^{\#} \text{ and } P^{\perp}_{\mathbf{K}^{-1/2}\mathbf{t}} = \mathbf{I} - P_{\mathbf{K}^{-1/2}\mathbf{t}} \quad (14)$$

where

$$\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^{\#} = \left(\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\left(\mathbf{K}^{-1/2}\mathbf{t}\right)\right)^{-1}\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T \quad (15)$$

becomes the well-known OSP matrix widely used in hyperspectral data exploitation [2], [3]. Substituting (14) into (13) yields (9), that is,

$$\begin{aligned}\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}) &= \left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right)^T P_{\mathbf{K}^{-1/2}\mathbf{t}}\left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right) \\ &= \left\|P_{\mathbf{K}^{-1/2}\mathbf{t}}\left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right)\right\|^2. \quad (16)\end{aligned}$$

■

Theorem 2: For any fixed data sample $\mathbf{r}$, we consider $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ as a function of its dual variable, the unknown target signature $\mathbf{t}$. Then we can find the optimal solution to (8) and (9), $\mathbf{t}^*$, given by

$$\mathbf{t}^* = \arg\left\{\max_{\mathbf{t}}\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})\right\} = \kappa(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \quad (17)$$

subject to any constant κ. The GAD can be derived by

$$\delta_{\kappa}^{\mathrm{GAD}}(\mathbf{r}) = \delta_{\mathbf{t}^*}^{\mathrm{GAD}}(\mathbf{r}) = \kappa(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}). \quad (18)$$

Specifically, by setting $\kappa = 1$ in (18), $\delta_{\kappa=1}^{\mathrm{GAD}}(\mathbf{r})$ turns out to exactly $\delta^{\mathrm{RXD}}(\mathbf{r})$ given by

$$\delta_{\kappa=1}^{\mathrm{GAD}}(\mathbf{r}) = (\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) = \delta^{\mathrm{RXD}}(\mathbf{r}). \quad (19)$$

Proof: In order to facilitate the derivation, we introduce a covariance-matrix K-whitened target signature $\widehat{\mathbf{t}}$ defined by

$$\widehat{\mathbf{t}} = \mathbf{K}^{-1/2}\mathbf{t}. \quad (20)$$

Then, $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ in (9) can be re-expressed

$$\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}) = \frac{\left\|\widehat{\mathbf{t}}^T\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right\|^2}{||\widehat{\mathbf{t}}||^2} = \left[\left(\frac{\widehat{\mathbf{t}}}{||\widehat{\mathbf{t}}||}\right)^T\frac{\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})}{||\widehat{\mathbf{t}}||}\right]. \quad (21)$$

By Schwarz's inequality, we obtain

$$\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}) \le \left\|\frac{\widehat{\mathbf{t}}}{||\widehat{\mathbf{t}}||}\right\|\left\|\frac{\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})}{||\widehat{\mathbf{t}}||}\right\| \quad (22)$$

where the equality in (22) holds if and only if

$$\frac{\widehat{\mathbf{t}}^*}{||\widehat{\mathbf{t}}^*||} = \kappa\frac{\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})}{||\widehat{\mathbf{t}}^*||} \Rightarrow \widehat{\mathbf{t}}^* = \kappa\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \quad (23)$$

which proves (17).

Alternatively, also by Schwarz's inequality, (17) can be further shown as

$$\begin{aligned}\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}) &= \left\|P_{\mathbf{K}^{-1/2}\mathbf{t}}\left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right)\right\|^2 \\ &= \left\|P_{\widehat{\mathbf{t}}}\left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right)\right\|^2 \le \left\|P_{\widehat{\mathbf{t}}}\right\|\left\|\left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right)\right\| \quad (24)\end{aligned}$$

where the equality in (24) holds if and only if

$$\begin{aligned}P_{\widehat{\mathbf{t}}^*} &= \kappa\left(\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right) \text{ for any constant } \kappa \\ &\Rightarrow \widehat{\mathbf{t}}^* = \kappa\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \Rightarrow \widehat{\mathbf{t}}^* = \kappa(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \quad (25)\end{aligned}$$

which is exactly the same as (23). ■

It is worth noting that since the hypothesis testing theory requires that the target signature be detected, $\mathbf{t}$ under the alternative hypothesis H_1 must be known. The key of Theorem 2 is to treat the target signature $\mathbf{t}$ used in $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ as the dual variable of the data sample $\mathbf{r}$ to serve as a dummy variable to play the role of liaison. As a consequence, $\mathbf{t}$ is eventually vanished by being replaced with its optimal value, $\mathbf{t}^*$ in (17) to derive $\delta^{\mathrm{GLRT}}(\mathbf{r})$ as a special case of $\delta_{\kappa}^{\mathrm{GAD}}(\mathbf{r})$ in (18) where $\delta_{\kappa=1}^{\mathrm{GAD}}(\mathbf{r}) = \delta^{\mathrm{RXD}}(\mathbf{r})$ in (19).

IV. Interpretation of GLRT by SNR

Using SNR to design a target detector has been discussed and studied extensively in [10]. For those who are interested in this theory, we refer to details in [10].

The following theorem links GLRT and SNR, in which case the two dual variables, the data sample $\mathbf{r}$ and target signature $\mathbf{t}$, play interchanged roles. In other words, the data sample $\mathbf{r}$ is now being given and fixed, while the target signature $\mathbf{t}$ is assumed to be an unknown signature variable.

Theorem 3: For each given and fixed data sample $\mathbf{r}$, RXD is the maximum SNR of $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ in (7) over the unknown target signature $\mathbf{t}$, that is,

$$\delta^{\mathrm{RXD}}(\mathbf{r}) = \max_{\mathbf{t}}\mathrm{SNR}\left(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})\right). \quad (26)$$

That is, $\delta_{\kappa=1}^{\mathrm{GAD}}(\mathbf{r}) = \delta^{\mathrm{RXD}}(\mathbf{r})$ in (19).

Proof: Now, we can find the maximum SNR of (9) over $\mathbf{t}$ as follows:

$$\begin{aligned}\mathrm{SNR}\left(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})\right) &= \frac{\left(E[\delta^{\mathrm{GLRT}}(\mathbf{r})|H_1] - E[\delta^{\mathrm{GLRT}}(\mathbf{r})|H_0]\right)^2}{\mathrm{var}[\delta^{\mathrm{GLRT}}(\mathbf{r})|H_0]} \\ &= \frac{(\hat{\alpha}_{\mathrm{MLE}}(\mathbf{r}))^4\left(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}\right)^2}{(\hat{\alpha}_{\mathrm{MLE}}(\mathbf{r}))^2\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}} = (\hat{\alpha}_{\mathrm{MLE}}(\mathbf{r}))^2\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t} \\ &= \left(\frac{(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}\mathbf{t}}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}}\right)^2\mathbf{t}^T\mathbf{K}_{L\times L}^{-1}\mathbf{t}\end{aligned}$$

which implies

$$\mathrm{SNR}\left(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})\right) = \frac{\left((\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}\mathbf{t}\right)^2}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}}. \quad (27)$$

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Equation (27) can be further simplified to

$$\begin{aligned}\mathrm{SNR}\big(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})\big) &= \frac{\left\|\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1/2}\mathbf{t}\right\|^2}{\left\|\mathbf{K}^{-1/2}\mathbf{t}\right\|^2}\\ &= \left\|\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\frac{\mathbf{K}^{-1/2}\mathbf{t}}{\left\|\mathbf{K}^{-1/2}\mathbf{t}\right\|}\right\|^2\\ &\le \left\|\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\right\|\left\|\frac{\mathbf{K}^{-1/2}\mathbf{t}}{||\mathbf{K}^{-1/2}\mathbf{t}||}\right\|\\ &= \left\|\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\right\|\left\|\frac{\widehat{\mathbf{t}}}{||\widehat{\mathbf{t}}||}\right\| \end{aligned} \tag{28}$$

with the equality if and only if

$$\frac{\widehat{\mathbf{t}}^*}{\left\|\widehat{\mathbf{t}}^*\right\|} = \kappa\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \tag{29}$$

where $\widehat{\mathbf{t}}^*$ is defined in (23).

Substituting (29) into (28) yields (26) and

$$\begin{aligned}\max_{\mathbf{t}}\mathrm{SNR}\big(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})\big) &= \delta_{\kappa=1}^{\mathrm{GAD}}(\mathbf{r}) = \left\|\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right\|^2\\ &= (\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \end{aligned} \tag{30}$$

which turns out to be RXD in (19) with $\kappa = 1$. ■

An interesting finding from Theorem 3 is noteworthy. Its idea is very similar to that of using the double maxima formulated by Blahut in [63, Th. 1] to calculate channel capacity [64], [65]. More specifically, the idea of modeling $\delta^{\mathrm{RXD}}(\mathbf{r})$ as the maximization problem of $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ over its dual target variable $\mathbf{t}$ in (26) is exactly the same as the idea of modeling mutual information defined by $I(X;Y) = \sum_{j=1}^{J} p_j \sum_{k=1}^{K} Q_{k|j}\log(Q_{k|j}/q_k)$ with the channel input space, channel output space and the forward channel specified by X with the probability vector $\mathbf{p} = (p_1, p_2, \ldots, p_J)^T$, Y with the probability vector $\mathbf{q} = (q_1, q_2, \ldots, q_k)^T$ and $\{Q_{k|j}\}_{j=1,k=1}^{J,K}$, respectively.

Specifically, if we introduce

$$J(\mathbf{p}, \hat{\mathbf{P}}) = \sum_{j=1}^{J} p_j \sum_{k=1}^{K} Q_{k|j}\log\frac{\hat{P}_{j|k}}{q_k}$$

where $\hat{\mathbf{P}} = \{\hat{P}_{j|k}\}_{k=1,j=1}^{K,J}$ is a backward channel, then mutual information $I(X,Y)$ can be now reformulated as the following maximization problem:

$$I(X,Y) = \max_{\hat{\mathbf{P}}}\big\{J(\mathbf{p}, \hat{\mathbf{P}})\big\} \tag{31}$$

over all possible backward channels $\hat{\mathbf{P}}$. The solution to (31), $\hat{\mathbf{P}}^*$, is given by $\hat{P}_{j|k}^* = (Q_{k|j}\,p_j/q_k^*)$ with $q_k^* = \sum_{j=1}^{J} p_j Q_{k|j}$. As a consequence, the dummy backward channel $\hat{\mathbf{P}}$ vanishes since its optimal solution $\hat{\mathbf{P}}^*$ is actually determined by the given forward channel $\{Q_{k|j}\}_{j=1,k=1}^{J,K}$ and channel input probability vector $\mathbf{p} = (p_1, p_2, \ldots, p_J)^T$. The maximum formulation in (31) turns out to be exactly the same as (26) where the pair of the input probability $\mathbf{p}$ and the backward channel $\hat{\mathbf{P}}$ in (31) can be considered as a set of dual vector variables similar to the set of dual variables, data sample, $\mathbf{r}$ and target signature, $\mathbf{t}$ used in GLRT in (26).

V. Signal-to-Noise Ratio Approach to Anomaly Detection

Since the probability density functions, $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$, used in $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ are generally unknown in many real applications, $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ cannot be applied. This is because if an anomalous target is of interest and needs to be detected, the "$\mathbf{t}$" in (1) is practically unknown. In this case, formulating AD as a binary hypothesis-testing problem with an assumed probability distribution under each hypothesis seems inappropriate.

Interestingly, according to Theorem 3, we can take another look at $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ in (7) by considering $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ as a function of two dual variables, $\mathbf{r}$ and $\mathbf{t}$. Now, for each given data sample $\mathbf{r}$, $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ can be treated as a function of its dual variable, target signature $\mathbf{t}$ which is an unknown parameter variable. Then, the maximum SNR of $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ in (7) over all possible target signature $\mathbf{t}$ turns out to be RXD proved by Theorem 3. This leads to a new direction deviating from the conventional hypothesis testing theory by adopting SNR as a criterion optimality for designing an optimal anomaly detection. Apparently, using SNR as a detection criterion has its own theory in [14] and is also known as deflection, $\mathrm{D}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}))$ in [66] defined by

$$\mathrm{D}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})) = \frac{\big(E[\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})|H_1] - E[\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})(\mathbf{r})|H_0]\big)^2}{\mathrm{var}[\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})|H_0]} \tag{32}$$

where $E[\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})|H_j]$ is expectation calculated by probability density function $p_j(\mathbf{r})$ under hypothesis H_j. However, (32) is exactly the same as (28) given by

$$\mathrm{D}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})) = \mathrm{SNR}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})) = \frac{1}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}}\big(\mathbf{t}^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\big)^2. \tag{33}$$

Accordingly, maximizing $\mathrm{D}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}))$ in (33) over $\mathbf{t}$ is equivalent to maximizing $\mathrm{SNR}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r}))$ over $\mathbf{t}$. That is,

$$\max_{\mathbf{t}}\mathrm{D}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})) = \max_{\mathbf{t}}\mathrm{SNR}(\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})) \tag{34}$$

which provides a base to use SNR as an alternative criterion to design an optimal detector without appealing for probability density functions, $p_0(\mathbf{r})$ and $p_1(\mathbf{r})$ used by GLRT.

This section presents a practical approach to designing an optimal anomaly detector using SNR via (26) as a criterion without using probability distributions assumed in (1). In particular, it is a much easier, simpler, and intuitive approach as an alternative to the approach proposed by Reed and Yu [12] which assumed that data were Gaussian distributed under each hypothesis and reference data must be also available to determine the unknown target signature $\mathbf{t}$.

Its idea is similar to that used to derived CEM in [5] which utilizes minimum variance (i.e., least squares error) as the criterion for optimality to design a finite impulse response (FIR) filter specified by an L-dimensional weighting vector $\mathbf{w}$ as

$$\delta_{\mathbf{w}}(\mathbf{r}) = \mathbf{w}^T\mathbf{r}. \tag{35}$$

where the detector δ operating on the original data space $\mathbf{X}$ with $\mathbf{r} \in \mathbf{X}$.

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As noted in the introduction, an anomaly detector is a passive detector without knowing the targets that it intends to detect compared to HTD which is an active detector and knows exactly what targets it attempts to detect. Since there is no prior knowledge about the targets to be detected, the three scenarios, pure pixel, subpixel, and mixed pixel target detection models discussed in [10] are not applicable to HAD. Accordingly, there is no need of estimating target abundance fractions such as MLE used in GLRT and Mahalanobis distance (MD) used in the SNR-theory in [10]. So, in this section, the SNR-based AD to be presented assumes that the target signature $\mathbf{t}$ used in (1) is not provided *a priori* and the only knowledge available is the data sample $\mathbf{r}$. In this case, the hypothesis-testing problem formulation specified by (1) is no longer applied to HAD. In addition, anomalies possess spectral properties of being relatively small, unexpected presence, and low probability occurrence which generally cannot be characterized by the second-order statistics and may be also very likely embedded in BKG. In order to bring anomalies out of BKG, three data whitening techniques developed in [10] are used for this purpose so that BKG can be either removed or suppressed through a whitening process.

A. Data Sphered-Based Anomaly Detection

The form of $\delta_{\mathbf{t}}^{\mathrm{GLRT}}(\mathbf{r})$ in (9) basically performs DS on the original data space via

$$\hat{\mathbf{r}} = \mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}) \tag{36}$$

to obtain a sphered data space, $\hat{\mathbf{X}}$ and the sphered data sample, $\hat{\mathbf{r}}$. As a result, $\hat{\mathbf{X}}$ has zero mean and identity matrix as its covariance matrix. In this case, the noise has been sphered and the noise term specified by the denominator in (32) has vanished. Consequently, SNR can be simply calculated by the filter output signal energy, $(\mathbf{w}^T\hat{\mathbf{r}})^2$ resulting from operating the linear filter (35) on $\hat{\mathbf{X}}$ as follows:

$$\mathrm{SNR}_{\mathbf{w}}(\delta_{\mathbf{w}}^{\mathrm{DS}}(\hat{\mathbf{r}})) = \frac{\left(\mathbf{w}^T\hat{\mathbf{r}}\right)^T\mathbf{w}^T\hat{\mathbf{r}}}{\left[E\left(\mathbf{w}^T\hat{\mathbf{b}}\right)^2\right]} = \frac{\left(\mathbf{w}^T\hat{\mathbf{r}}\right)^2}{\left[\mathbf{w}^T\mathbf{w}\right]} = \left(\bar{\mathbf{w}}^T\hat{\mathbf{r}}\right)^2 \le \hat{\mathbf{r}}^T\hat{\mathbf{r}} \tag{37}$$

where $\bar{\mathbf{w}} = (\mathbf{w}/||\mathbf{w}||)$ with $||\mathbf{w}||^2 = \mathbf{w}^T\mathbf{w}$ and the equality in (37) holds if and only if

$$\mathbf{w}^* = \kappa\hat{\mathbf{r}} \tag{38}$$

for any constant κ by Schwarz's inequality.

Substituting (38) back into (37) yields

$$\begin{aligned}\delta_{\kappa}^{\text{DS-AD}}(\mathbf{r}) &= \left(\mathbf{w}^*\right)^T\hat{\mathbf{r}} \\ &= \kappa\left[\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right]^T\left[\mathbf{K}^{-1/2}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})\right] \\ &= \kappa(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}).\end{aligned} \tag{39}$$

Since (39) is designed for AD, it is free of any arbitrary constant κ. So, we can let $\kappa = 1$. Then, $\delta_{\kappa=1}^{\text{DS-AD}}(\mathbf{r})$ in (39) becomes exactly the same as the detector, RXD in (19) developed in [12].

B. Covariance Matrix-Whitened Anomaly Detection

Following a similar approach to designing an optimal anomaly detector on $\hat{\mathbf{X}}$ in Section V-A, we also perform data whitening on the original data space using the data whitening matrix $\mathbf{K}^{-1/2}$ to obtain the $\mathbf{K}^{-1/2}$-whitened data space, $\widehat{\mathbf{X}}$ and data sample, $\widehat{\mathbf{r}}$ via

$$\widehat{\mathbf{r}} = \mathbf{K}^{-1/2}\mathbf{r}. \tag{40}$$

It should be noted that (40) is different from (36) in the sense that $\widehat{\mathbf{X}}$ is uncorrelated by $\mathbf{K}^{-1/2}$ with variances normalized to one but does not necessarily have zero mean. Then, we operate (35) on the $\mathbf{K}^{-1/2}$-whitened space, $\widehat{\mathbf{X}}$ to calculate the SNR from its filter output energy, $(\mathbf{w}^T\widehat{\mathbf{r}})^2$, given by

$$\begin{aligned}\mathrm{SNR}(\delta_{\mathbf{w}}^{\text{K-space}}(\widehat{\mathbf{r}})) &= \frac{\left(\mathbf{w}^T\widehat{\mathbf{r}}\right)^T\mathbf{w}^T\widehat{\mathbf{r}}}{\left[E\left(\mathbf{w}^T\widehat{\mathbf{b}}\right)^2\right]} = \frac{\left(\mathbf{w}^T\widehat{\mathbf{r}}\right)^2}{\left[\mathbf{w}^T\mathbf{w}\right]} \\ &= \left(\bar{\mathbf{w}}^T\widehat{\mathbf{r}}\right) \le \widehat{\mathbf{r}}^T\widehat{\mathbf{r}}\end{aligned} \tag{41}$$

where $\bar{\mathbf{w}} = (\mathbf{w}/||\mathbf{w}||)$ with $||\mathbf{w}||^2 = \mathbf{w}^T\mathbf{w}$ and the equality in (41) holds if and only if

$$\mathbf{w}^* = \kappa\widehat{\mathbf{r}} \tag{42}$$

for any constant κ by Schwarz's inequality.

Substituting (42) back into (41) yields

$$\delta_{\kappa}^{\text{K-AD}}(\mathbf{r}) = \left(\mathbf{w}^*\right)^T\widehat{\mathbf{r}} = \kappa\left[\mathbf{K}^{-1/2}\mathbf{r}\right]^T\left[\mathbf{K}^{-1/2}\mathbf{r}\right] = \kappa\mathbf{r}^T\mathbf{K}^{-1}\mathbf{r}. \tag{43}$$

Since (35) is designed for AD, it is independent of any arbitrary constant κ. So, we can let $\kappa = 1$. Then, $\delta_{\kappa=1}^{\text{K-AD}}(\mathbf{r})$ in (43) becomes

$$\delta_{\kappa=1}^{\text{K-AD}}(\mathbf{r}) = \mathbf{r}^T\mathbf{K}^{-1}\mathbf{r}. \tag{44}$$

which can be considered as K-correlator or K-correlated filter because the sample $\mathbf{r}$ correlates itself via the sample covariance matrix $\mathbf{K}$ to make $\delta_{\kappa=1}^{\text{K-AD}}(\mathbf{r})$ a quadratic filter.

C. Correlation Matrix-Whitened Anomaly Detection

Once again, we can also perform data whitening on the original data space by using $\mathbf{R}^{-1/2}$ to obtain the $\mathbf{R}^{-1/2}$-whtened data space, $\bar{\mathbf{X}}$ and data sample, $\bar{\mathbf{r}}$ via

$$\bar{\mathbf{r}} = \mathbf{R}^{-1/2}\mathbf{r} \tag{45}$$

where the noise variance has been normalized to $\sigma_{\bar{\mathbf{b}}}^{-2}$.

Now, we operate (35) on the $\mathbf{R}^{-1/2}$-whitened space, $\bar{\mathbf{X}}$ to find the SNR from its linear output energy, $(\mathbf{w}^T\bar{\mathbf{r}})^2$

$$\begin{aligned}\mathrm{SNR}(\delta_{\mathbf{w}}^{\text{R-space}}(\bar{\mathbf{r}})) &= \frac{\left(\mathbf{w}^T\bar{\mathbf{r}}\right)^T\mathbf{w}^T\bar{\mathbf{r}}}{\left[E\left(\mathbf{w}^T\bar{\mathbf{b}}\right)^2\right]} = \frac{\left(\mathbf{w}^T\bar{\mathbf{r}}\right)^2}{\left[\sigma_{\bar{\mathbf{b}}}^2\mathbf{w}^T\mathbf{w}\right]} \\ &= \sigma_{\bar{\mathbf{b}}}^{-2}\left(\bar{\mathbf{w}}^T\bar{\mathbf{r}}\right) \le \sigma_{\bar{\mathbf{b}}}^{-2}\bar{\mathbf{r}}^T\bar{\mathbf{r}}\end{aligned} \tag{46}$$

where $\bar{\mathbf{w}} = (\mathbf{w}/||\mathbf{w}||)$ and the equality in (39) holds if and only if

$$\mathbf{w}^* = \kappa\bar{\mathbf{r}} \tag{47}$$

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TABLE I
SNR-DERIVED ANOMALY DETECTORS

SNR-based anomaly detectors	$\kappa = 1$	equation
DS-anomaly detector (DS-AD) $\delta_{\kappa}^{\text{DS-AD}}(\mathbf{r}) = \kappa\left(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}\right)^{T}\mathbf{K}^{-1}\left(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}\right)$	$\delta^{\text{RXD}}(\mathbf{r}) = \left(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}\right)^{T}\mathbf{K}^{-1}\left(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}\right)$	(39)/(19)
K-anomaly detector (K-AD) $\delta_{\kappa}^{\text{K-AD}}(\mathbf{r}) = \kappa\mathbf{r}^{T}\mathbf{K}^{-1}\mathbf{r}$	$\delta_{\kappa=1}^{\text{K-AD}}(\mathbf{r}) = \mathbf{r}^{T}\mathbf{K}^{-1}\mathbf{r}$	(43)/(44)
R-anomaly detector (R-AD) $\delta_{\kappa}^{\text{R-AD}}(\mathbf{r}) = \kappa\mathbf{r}^{T}\mathbf{R}^{-1}\mathbf{r}$	$\delta^{\text{R-AD}}(\mathbf{r}) = \mathbf{r}^{T}\mathbf{R}^{-1}\mathbf{r}$	(48)/(49)

for any constant κ by Schwarz's inequality.

Substituting (47) into (46) yields

$$\delta_{\kappa}^{\text{R-AD}}(\mathbf{r}) = \left(\mathbf{w}^{*}\right)^{T}\bar{\mathbf{r}} = \kappa\left[\mathbf{R}^{-1/2}\mathbf{r}\right]^{T}\left[\mathbf{R}^{-1/2}\mathbf{r}\right] = \kappa\mathbf{r}^{T}\mathbf{R}^{-1}\mathbf{r}. \quad (48)$$

Now, let $\kappa = 1$, $\delta_{\kappa=1}^{\text{R-AD}}(\mathbf{r})$ in (48) turns out to be the detector

$$\delta_{\kappa=1}^{\text{R-AD}}(\mathbf{r}) = \mathbf{r}^{T}\mathbf{R}^{-1}\mathbf{r} \quad (49)$$

called R-anomaly detector (R-AD) developed in [67] where no theory was provided. In correspondence to (44), R-AD in (49) can be considered R-correlator or R-correlated filter because the sample $\mathbf{r}$ correlates itself via the sample correlation matrix $\mathbf{R}$ to make $\delta_{\kappa=1}^{\text{R-AD}}(\mathbf{r})$ a quadratic filter.

VI. RELATIONSHIP BETWEEN LRT—AND SNR-DERIVED ANOMALY DETECTORS

The relationship between GLRT-derived anomaly detectors in Section III and SNR-derived anomaly detectors in Section V is established by Theorem 3 in Section IV and the deflection criterion specified by (32) which is essentially SNR. There are several key differences between them. One is their problem formulations. GLRT is an optimal filter resulting from a hypothesis-testing problem specified by (1). Compared to GLRT, SNR-based anomaly detectors are derived from linear filters specified by (35) using SNR as a criterion for optimality. Another is the requirement of a probability distribution under each hypothesis, specifically, Gaussian distributions so that GLRT can be calculated. By contrast, the SNR-based HAD requires only the first 2 orders of data statistics without assuming probability distributions. Particularly, the threshold τ used by GLRT in (3) and (4) and (7) as a parameter must be determined by costs and prior probability distributions formulated by hypothesis-testing problems corresponding κ used in SNR which can be determined by applications. As a matter of fact, it is the parameter κ which makes SNR-approach more interesting and appealing than the LRT-based approach. In order to see that, Table I summarizes SNR-derived anomaly detectors along with the parameter κ set to $\kappa = 1$.

VII. HAD: DUAL THEORY OF HTD

Despite the fact that HTD and HAD are considered as active and passive target detections, they can be considered as dual concepts, both of which perform MFs with the critical difference in how to use their matched signals. It is this matched signal to make HTD is an active detector, while making HAD a passive detector as noted in the introduction. In HTD, the matched signal is specified by *a priori* target signature $\mathbf{t}$ as opposed to HAD whose matched signal is specified by the data sample $\mathbf{r}$ itself currently being processed, which is unknown and varies data sample-by-data sample.

A. DS-TD as DS-AD

In this section, we can derive a dual theorem corresponding to Theorem 2 of Theorem 1 for HTD as follows:

Theorem 4: For any fixed target signature $\mathbf{t}$, consider $\delta_{\mathbf{t}}^{\text{GLRT}}(\mathbf{r})$ in (8) and (9) as a function of its dual variable, the data sample $\mathbf{r}$. Then, the optimal solution to (8) and (9), $\mathbf{r}^{*}$, can be found by

$$\mathbf{r}^{*} - \boldsymbol{\mu}_{\mathbf{b}} = \arg\left\{\max_{\mathbf{r}} \delta_{\mathbf{t}}^{\text{GLRT}}(\mathbf{r})\right\} = \kappa\mathbf{t} \quad (50)$$

with κ being any constant. As a result of (50), HTD can be derived as

$$\delta_{\kappa}^{\text{GLRT}}(\mathbf{r}) = \delta_{\mathbf{r}^{*}}^{\text{GLRT}}(\mathbf{r}) = \kappa\mathbf{t}^{T}\mathbf{K}^{-1}\mathbf{t} \quad (51)$$

which is the maximum SNR of GLRT without κ as shown in [10].

Proof: According to (9)

$$\begin{aligned}\delta_{\mathbf{t}}^{\text{GLRT}}(\mathbf{r}) &= \frac{\left\|\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^{T}\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right\|^{2}}{||\mathbf{K}^{-1/2}\mathbf{t}||^{2}} \\ &\le \left\|\frac{\mathbf{K}^{-1/2}\mathbf{t}}{||\mathbf{K}^{-1/2}\mathbf{t}||}\right\|\left\|\frac{\mathbf{K}^{-1/2}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})}{||\mathbf{K}^{-1/2}\mathbf{t}||}\right\| \end{aligned} \quad (52)$$

where the equality holds if and only if

$$\mathbf{r}^{*} - \boldsymbol{\mu}_{\mathbf{b}} = \kappa\mathbf{t}. \quad (53)$$

Substituting (53) into (52) yields

$$\begin{aligned}\max_{\mathbf{r}} \delta_{\mathbf{t}}^{\text{GLRT}}(\mathbf{r}) &= \delta_{\mathbf{t}}^{\text{GLRT}}(\mathbf{r}^{*}) \\ &= \kappa\left(\mathbf{K}^{-1/2}\mathbf{t}\right)^{T}\left(\mathbf{K}^{-1/2}\mathbf{t}\right) = \kappa\mathbf{t}^{T}\mathbf{K}^{-1}\mathbf{t}. \end{aligned} \quad (54)$$

■

Interestingly, if $\mathbf{r} - \boldsymbol{\mu}_{\mathbf{b}}$ in (8) is replaced by $\mathbf{t}$, (8) becomes (54) subject to the constant κ. On the other hand, if the second $\mathbf{t}$ in (54) is replaced by $\mathbf{r}^{*} - \boldsymbol{\mu}_{\mathbf{b}}$, (54) becomes

$$\delta_{\kappa}^{\text{GLRT}}(\mathbf{r}) = \kappa(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^{T}\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \quad (55)$$

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subject to the constant κ. In particular, when $\kappa = 1$, (55) turns out to

$$\delta^{\mathrm{GLRT}}_{\kappa=1}(\mathbf{r}) = \delta^{\mathrm{AMD}}(\mathbf{r}) = (\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}) \tag{56}$$

which is called AMD developed in [10] corresponding to AMF in [16] with its normalization referred to as AMD (NAMD) also derived in [10]. It should be noted that if $\mathbf{t}$ and $\mathbf{r}$ in (56) are identical, (56) becomes MD. In particular, two scenarios can be derived from (56).

1) If $\mathbf{t}$ in (56) is replaced by $\mathbf{r}$, the resulting AMD with $\mathbf{t} \leftarrow \mathbf{r}$ becomes RXD in (19) and also (31).
2) If we interchange the roles of $\mathbf{t}$ and $\mathbf{r}$ by replacing $\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}}$ in (56) with $\mathbf{t}$, the resulting AMD with $\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}} \leftarrow \mathbf{t}$ becomes (54) with $\kappa = 1$ and also results in $\delta^{\mathrm{GLRT}}_{\mathbf{t}}(\mathbf{r}^*) = \max_{\mathbf{r}}\mathrm{SNR}(\delta^{\mathrm{GLRT}}_{\mathbf{t}}(\mathbf{r}))$ with $\mathbf{r}^*-\boldsymbol{\mu}_{\mathbf{b}} = \mathbf{t}$ derived in [10].

Specifically, NAMD can be obtained from (39) and (55) by setting $\kappa = [(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})]^{-1}$ in (39) and (55) to yield

$$\begin{aligned}\delta^{\mathrm{DS\text{-}AD}}_{\kappa=(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})}(\mathbf{r}) &= \delta^{\mathrm{GLRT}}_{\kappa=(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})}(\mathbf{r}) \\ &= \frac{(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})}{(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})} = \delta^{\mathrm{NAMD}}(\mathbf{r}).\end{aligned} \tag{57}$$

B. K-TD as K-AD

Using an idea similar to Corollary 2, we can even extend the traditional hypothesis testing-based signal detection problem to an AD problem as follows:

$$\begin{aligned} &H_0 : \mathbf{r} = \mathbf{b} \\ &\text{versus} \\ &H_1 : \mathbf{r} = \mathbf{t} + \mathbf{b} \end{aligned} \tag{58}$$

where $\mathbf{r}$ and $\mathbf{t}$ are the data sample and target signal with $\mathbf{b}$ being the BKG signal. It should be noted that $\mathbf{t}$ in (58) is known perfectly with no the target abundance fraction α as formulated in (1). The optimal target detector can be obtained by an LRT [10], [14] as

$$\delta^{\mathrm{LRT}}_{\mathbf{t}}(\mathbf{r}) = \begin{cases} 1, & \text{if } \mathbf{t}^T\mathbf{K}^{-1}\mathbf{r} \geq \tau' \\ 0, & \text{if } \mathbf{t}^T\mathbf{K}^{-1}\mathbf{r} \geq \tau' \end{cases} \tag{59}$$

where $\delta^{\mathrm{LRT}}_{\mathbf{t}}(\mathbf{r})$ is particularly expressed as a function of $\mathbf{r}$ and $\mathbf{t}$. Now, we redefine $\delta^{\mathrm{LRT}}_{\mathbf{t}}(\mathbf{r})$ as $\delta^{\mathrm{K\text{-}TD}}_{\widehat{\mathbf{t}}}(\widehat{\mathbf{r}})$ to be a target detector operating in the K-space using (40) as

$$\begin{aligned}\delta^{\mathrm{K\text{-}TD}}_{\widehat{\mathbf{t}}}(\widehat{\mathbf{r}}) &= \widehat{\mathbf{t}}^T\widehat{\mathbf{r}} = \mathbf{t}^T\mathbf{K}^{-1}\mathbf{r} \\ &= \left(\mathbf{K}^{-1/2}\mathbf{t}\right)^T\left(\mathbf{K}^{-1/2}\mathbf{t}\right) \leq \left\|\mathbf{K}^{-1/2}\mathbf{t}\right\|\left\|\mathbf{K}^{-1/2}\mathbf{r}\right\|.\end{aligned} \tag{60}$$

There are two scenarios to make the equality in (60) true.

1) Assume that $\mathbf{t}$ is known and given. Then the data sample $\mathbf{r}^*$ makes the equality hold if and only if

$$\mathbf{K}^{-1/2}\mathbf{r}^* = \kappa\mathbf{K}^{-1/2}\mathbf{t} \Rightarrow \mathbf{r}^* = \kappa\mathbf{t} \text{ subject to the constant } \kappa \tag{61}$$

and

$$\max_{\widehat{\mathbf{r}}}\delta^{\mathrm{K\text{-}TD}}_{\widehat{\mathbf{t}}}(\widehat{\mathbf{r}}) = \kappa\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}. \tag{62}$$

a) Setting $\kappa = 1$ in (62) yields

$$\kappa = \left(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}\right)^{-1}. \tag{63}$$

By substituting (63) into (61), we obtain the following normalized LRT (NLRT) in [10]

$$\delta^{\mathrm{NLRT}}_{\mathbf{t}}(\mathbf{r}) = \frac{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{r}}{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}}. \tag{64}$$

2) Assume that $\mathbf{t}$ is unknown. Then, the target signature $\mathbf{t}^*$ makes the equality hold if and only if

$$\mathbf{K}^{-1/2}\mathbf{t}^* = \kappa\mathbf{K}^{-1/2}\mathbf{r} \Rightarrow \mathbf{t}^* = \kappa\mathbf{r}. \tag{65}$$

By virtue of (64), we can make $\delta^{\mathrm{K\text{-}TD}}_{\mathbf{t}}(\mathbf{r})$ in (62) an anomaly detector, $\delta^{\mathrm{K\text{-}AD}}_{\kappa}(\mathbf{r})$ by

$$\delta^{\mathrm{K\text{-}AD}}_{\kappa}(\mathbf{r}) = \max_{\mathbf{t}}\delta^{\mathrm{LRT}}_{\mathbf{t}}(\mathbf{r}) = \kappa\mathbf{r}^T\mathbf{K}^{-1}\mathbf{r}. \tag{66}$$

Setting $\kappa = 1$ in (66) yields

$$\delta^{\mathrm{K\text{-}AD}}_{\kappa=1}(\mathbf{r}) = \mathbf{r}^T\mathbf{K}^{-1}\mathbf{r} \tag{67}$$

which is exactly the K-AD derived from SNR in (44).

C. R-TD as R-AD

Unfortunately, there is no corresponding counterpart using sample correlation matrix $\mathbf{R}$ to derive LRT in [10] and [20], [22]. However, according to [10], there is an SNR-derived target detector using $\mathbf{R}$ given by

$$\delta^{\mathrm{R\text{-}SNR}}_{\kappa}(\mathbf{r}) = \kappa\bar{\mathbf{t}}^T\bar{\mathbf{r}} = \kappa\left(\mathbf{R}^{-1/2}\mathbf{t}\right)^T\left(\mathbf{R}^{-1/2}\mathbf{t}\right) = \kappa\mathbf{t}^T\mathbf{R}^{-1}\mathbf{r}. \tag{68}$$

Now, if we redefine $\delta^{\mathrm{R\text{-}SNR}}_{\kappa}(\mathbf{r})$ as $\delta^{\mathrm{R\text{-}TD}}_{\mathbf{t}}(\bar{\mathbf{r}})$ to a target detector by

$$\delta^{\mathrm{R\text{-}TD}}_{\kappa}(\bar{\mathbf{r}}) = \bar{\mathbf{t}}^T\bar{\mathbf{r}} = \kappa\left(\mathbf{R}^{-1/2}\mathbf{t}\right)^T\left(\mathbf{R}^{-1/2}\mathbf{t}\right) \leq \kappa\left\|\mathbf{R}^{-1/2}\mathbf{t}\right\|\left\|\mathbf{R}^{-1/2}\mathbf{r}\right\|. \tag{69}$$

In analogy with K-TD, there are two scenarios to make the equality in (69) true.

1) Assume that $\mathbf{t}$ is known and given. Then the data sample $\mathbf{r}^*$ makes the equality hold if and only if

$$\mathbf{R}^{-1/2}\mathbf{r}^* = \kappa\mathbf{R}^{-1/2}\mathbf{t} \Rightarrow \mathbf{r}^* = \kappa\mathbf{t}. \tag{70}$$

By substituting (70) into (69), we can derive an R-TD

$$\max \mathrm{SNR}\delta^{\mathrm{R\text{-}TD}}_{\kappa}(\mathbf{r}) = \mathbf{t}^T\mathbf{R}^{-1}\mathbf{t}. \tag{71}$$

a) Setting $\kappa = 1$ in (69) with equality yields

$$\delta^{\mathrm{R\text{-}TD}}_{\kappa=1}(\mathbf{r}) = \mathbf{t}^T\mathbf{R}^{-1}\mathbf{r} \tag{72}$$

which turns out to be the maximum SNR of R-TD in (69).

b) Making $\kappa = 1$ in (69), we can obtain κ also given by (63).
By substituting (63) into (69), we derive R-TD

$$\delta^{\mathrm{R\text{-}TD}}_{\kappa=\left(\mathbf{t}^T\mathbf{R}^{-1}\mathbf{t}\right)^{-1}}(\mathbf{r}) = \frac{\mathbf{t}^T\mathbf{R}^{-1}\mathbf{r}}{\mathbf{t}^T\mathbf{R}^{-1}\mathbf{t}} = \delta^{\mathrm{CEM}}(\mathbf{r}) \tag{73}$$

which turns out to be CEM [5], [6].

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TABLE II
RELATIONSHIP BETWEEN DS-DERIVED ANOMALY DETECTOR AND GLRT-DERIVED ANOMALY DETECTORS

SNR-derived anomaly detectors	GLRT-derived detectors	equation
DS-AD $\delta_{\kappa}^{\text{DS-AD}}(\mathbf{r}) = \kappa(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T \mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})$ in(39) and $\delta_{\kappa}^{\text{GLRT}}(\mathbf{r})$ in (51)	κ=1 $\delta^{\text{AMD}}(\mathbf{r}) = (\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T \mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})$	(56)
	$\kappa = \left((\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T \mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})\right)^{-1}$ $\delta^{\text{NAMD}}(\mathbf{r}) = \frac{(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T \mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})}{(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T \mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})}$	(57)
K- anomaly detector $\delta_{\kappa}^{\text{K-AD}}(\mathbf{r}) = \kappa \mathbf{r}^T \mathbf{K}^{-1}\mathbf{r}$ in (43)/(66)	κ=1 $\delta^{\text{LRT}}(\mathbf{r}) = \mathbf{t}^T \mathbf{K}^{-1}\mathbf{r}$	(44)=(67)
	$\kappa = (\mathbf{t}^T \mathbf{K}^{-1}\mathbf{t})^{-1}$ $\delta^{\text{NLRT}}(\mathbf{r}) = \frac{\mathbf{t}^T \mathbf{K}^{-1}\mathbf{r}}{\mathbf{t}^T \mathbf{K}^{-1}\mathbf{t}}$	(64)
R-anomaly detector $\delta_{\kappa}^{\text{R-AD}}(\mathbf{r}) = \kappa \mathbf{r}^T \mathbf{R}^{-1}\mathbf{r}$ in (48)/(68)	$\kappa = 1$ $\delta^{\text{R-SNR}}(\mathbf{r}) = \mathbf{t}^T \mathbf{R}^{-1}\mathbf{r}$	(49)/(72)
	$\kappa = (\mathbf{t}^T \mathbf{R}^{-1}\mathbf{t})^{-1}$ $\delta^{\text{CEM}}(\mathbf{r}) = \frac{\mathbf{t}^T \mathbf{R}^{-1}\mathbf{r}}{\mathbf{t}^T \mathbf{R}^{-1}\mathbf{t}}$	(73)

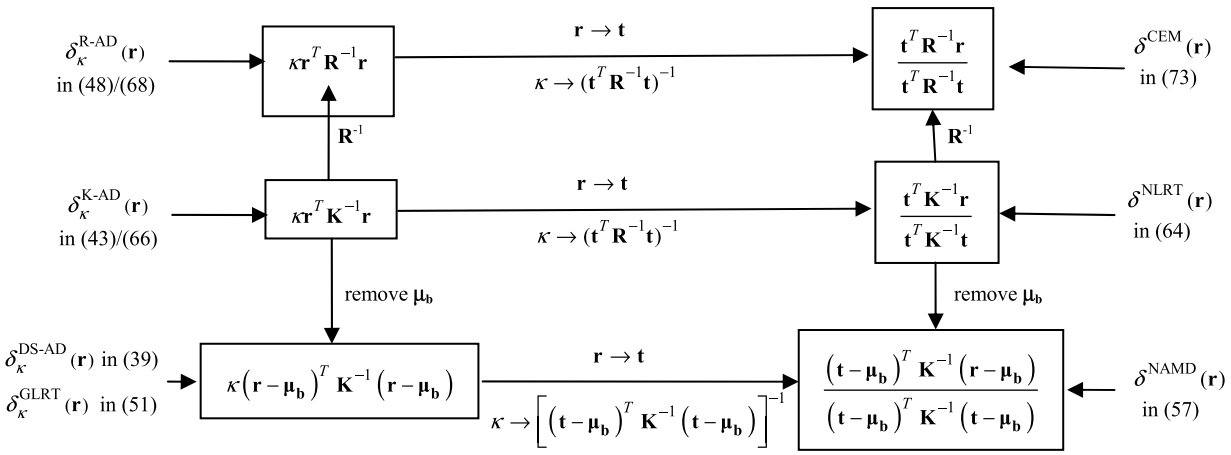


Fig. 1. Schematic of SNR-based HTD, NAMD, NLRT, and CEM, and their dual counterparts of the normalized versions of DS-AD, K-AD, and R-AD in HAD.

2) Assume that $\mathbf{t}$ is unknown. Then, the target signature $\mathbf{t}^*$ makes the equality hold if and only if

$$\mathbf{R}^{-1/2}\mathbf{t}^* = \kappa \mathbf{R}^{-1/2}\mathbf{r} \Rightarrow \mathbf{t}^* = \kappa \mathbf{r}. \tag{74}$$

By substituting (74) into (69), we can obtain an anomaly detector operating R-space as $\delta_{\kappa}^{\text{R-AD}}(\mathbf{r})$ given by

$$\delta_{\kappa}^{\text{R-AD}}(\mathbf{r}) = \kappa \mathbf{r}^T \mathbf{R}^{-1}\mathbf{r}. \tag{75}$$

Setting κ in (75) to 1 gives rise to

$$\delta_{\kappa=1}^{\text{R-AD}}(\mathbf{r}) = \mathbf{r}^T \mathbf{R}^{-1}\mathbf{r}. \tag{76}$$

Table II describes the relationship between DS-derived anomaly detectors and GLRT-derived anomaly detectors. A dual theory of SNR-based HTD developed in [10] was also shown in Section VII, where NAMD, NLRT, and CEM were the dual counterparts of the normalized versions of DS-AD, K-AD, and R-AD in HAD listed Table II and described in Fig. 1.

VIII. 3-D ROC CURVE-DERIVED DETECTION MEASURES

In HAD, a commonly used tool is the 2-D ROC curve which is a plot of detection probability P_{D} versus false alarm probability P_{F}. Its AUC, denoted by $\text{AUC}_{(\text{D,F})}$, is calculated as an indicator to measure how effective a detector is. Unfortunately, according to a recent work in [23], solely relying on $\text{AUC}_{(\text{D,F})}$ is not creditable. This is because both P_{D} and P_{F} are calculated by the same threshold τ used by the detector in (3). As a result, when both P_{D} and P_{F} are very high, its calculated $\text{AUC}_{(\text{D,F})}$ is also very high. Similarly, when both P_{D} and P_{F} are very low, its calculated $\text{AUC}_{(\text{D,F})}$ is also very low. In other

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words, P_D and P_F interact each other simultaneously and cannot work independently. Because of that, P_D and P_F cannot individually measure TD and BKG suppressibility (BS), respectively. To address this issue, Chang [23] developed an effective 3-D ROC analysis-based evaluation tool, which extends the traditional 2-D ROC analysis by including the threshold τ as an additional parameter to represent a 3-D RIOC curve as a function of three parameters, P_D, P_F, and τ. As a result, a 3-D ROC curve can be generated by a triplet parameter vector specified by (P_D,P_F,τ) and it can be also used to three 2-D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) with their respective AUC values, denoted by, $\text{AUC}_{(D,F)}$, $\text{AUC}_{(D,\tau)}$, and $\text{AUC}_{(F,\tau)}$, where $\text{AUC}_{(D,\tau)}$ and $\text{AUC}_{(F,\tau)}$ can be used to evaluate TD and BS, respectively. In addition to these three AUC values, five new AUC measures are also developed in [23] to measure joint TD, joint BS, TDBS, SNPR, and ODP defined in the following:

1) $\text{AUC}_{(D,F)}$*:* effectiveness of a detector;
2) $\text{AUC}_{(D,\tau)}$*:* TD of a detector;
3) $\text{AUC}_{(F,\tau)}$*:* BS of a detector;
4) AUC_{TD}*:* joint TD of a detector defined by

$$0 \le \text{AUC}_{TD} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)}) \le 2. \quad (77)$$

5) AUC_{BS}*:* joint BS of detector defined by

$$-1 \le \text{AUC}_{BS} = \text{AUC}_{(D,F)} - \text{AUC}_{(F,\tau)}) \le 1. \quad (78)$$

6) AUC_{TDBS}*:* TDBS of a detector defined by

$$-1 \le \text{AUC}_{TDBS} = \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)}) \le 1. \quad (79)$$

7) AUC_{SNPR}*:* SNPR of a detector defined by

$$0 \le \text{AUC}_{SNPR} = \frac{\text{AUC}_{(D,\tau)}}{\text{AUC}_{(F,\tau)}}. \quad (80)$$

8) AUC_{ODP}: ODP of a detector defined by

$$-1 \le \text{AUC}_{ODP} = \text{AUC}_{(D,F)} + \text{AUC}_{(D,\tau)} - \text{AUC}_{(F,\tau)} \le 2. \quad (81)$$

As a summary, the above detection measures can be generally categorized into the following three functionalities:

1) TD: $\text{AUC}_{(D,\tau)}$, AUC_{TD};
2) BS: $\text{AUC}_{(F,\tau)}$, AUC_{BS}, and AUC_{SNPR};
3) Detector's Effectiveness: $\text{AUC}_{(D,F)}$, AUC_{TDBS}, and AUC_{ODP}.

For those who are interested in using the 3-D ROC analysis developed in [23] for detection performance evaluation, a software package of calculating the above detection measures is available on the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL), University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, https://wiki.umbc.edu/display/rssipl/10.+Download [68].

IX. Images to Be Studied

Three hyperspectral real image scenes were used for experiments to illustrate three anomaly detectors, DS-AD, K-AD, and R-AD in Table II. It should be noted that DS-AD implemented in the experiments was the same as RXD.

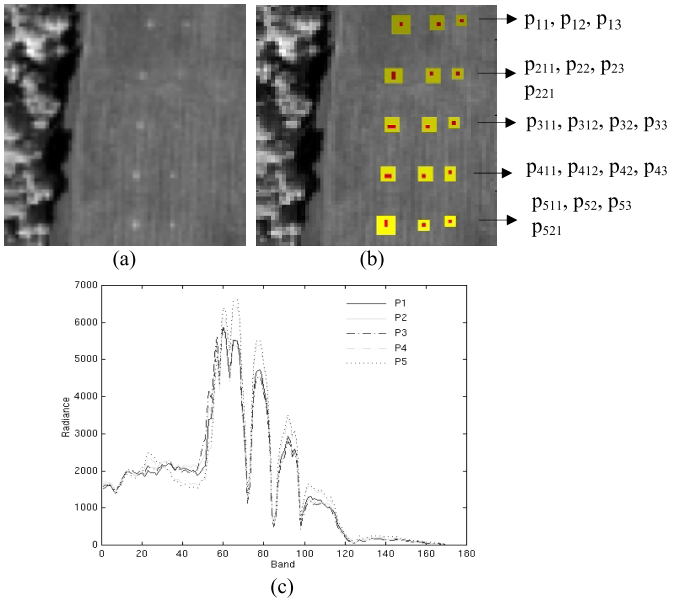


Fig. 2. (a) HYDICE panel scene which contains 15 panels. (b) Ground truth map of spatial locations of 19 R panel pixels. (c) Spectra of $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$.

A. HYDICE Panel Scene

An airborne HYDICE scene is shown in Fig. 2. It was collected in August 1995 from a flight altitude of 10 000 ft with the ground sampling distance approximately 1.56 m. This scene has been studied extensively by many reports such as [1] and [3]. It has a total of 169 bands which were used for the experiment with low signal/high noise bands: bands 1–3 and bands 202–210; and water vapor absorption bands: bands 101–112 and bands 137–153 removed. There are 15 square panels in Fig. 2(a) with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m, respectively. Due to the ground sampling distance approximately 1.56 m, the each of panels in the first column except the first row contains two panel pixels highlighted by red, p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 2. All the remaining 11 panels in Fig. 2 contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} andp_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 2(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. Fig. 2(c) plots the five-panel spectral signatures, $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$ obtained from Fig. 2(b), where the ith panel signature, denoted by $\mathbf{p}_i$ was generated by averaging the red panel center pixels in row i.

B. HYDICE Urban Scene

A second dataset is another HYDICE image with pseudo-color shown in Fig. 3(a). It is an urban scene and comprised 210 spectral bands with 174 bands being used for experiments after the noise and water absorption bands had been removed. A region with a size of 80 × 100 pixels located at the upper right of the scene was selected as the test image

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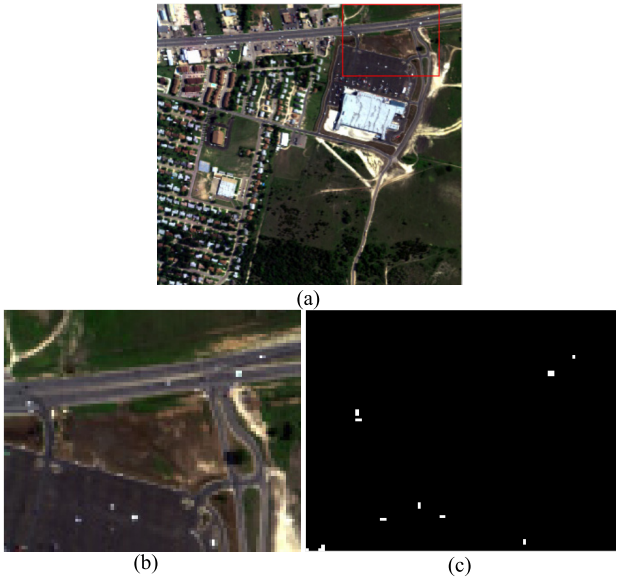


Fig. 3. HYDICE urban scene (a) pseudo-color image of the whole scene, (b) pseudo-color image of the selected area, and (c) ground truth map.

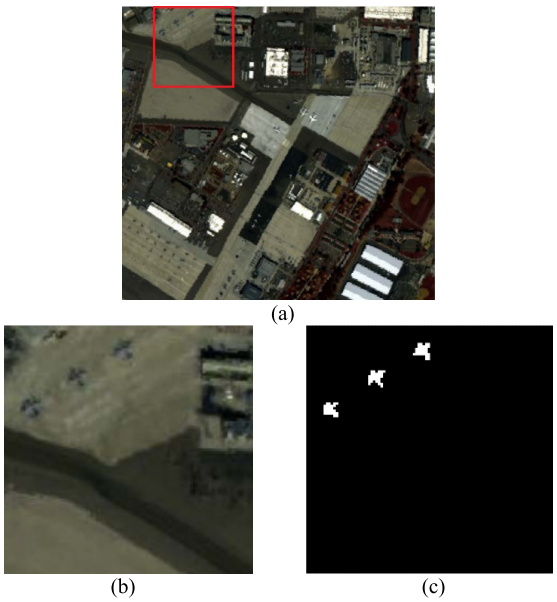


Fig. 4. San Diego airport scene (a) pseudo-color image of the whole scene, (b) pseudo-color image of the selected area, and (c) ground truth map.

shown in Fig. 3(b) along with the ground-truth map shown in Fig. 3(c) where 21 pixels were identified as anomalies, which were cars and roofs, because they had spectra that differ from BKG.

C. San Diego Airport Data

A third dataset used for experiments was also an airborne visible/infrared imaging spectrometer (AVIRIS) acquired scene, which is a San Diego airport scene in CA, USA with pseudo-color shown in Fig. 4(a). It has a size of 400 × 400 pixels with a 3.5-m spatial resolution and 224 spectral channels in wavelengths ranging from 370 to 2510 nm. After removing the bands that correspond to the water absorption regions, low SNR, and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 available bands of the data were retained in

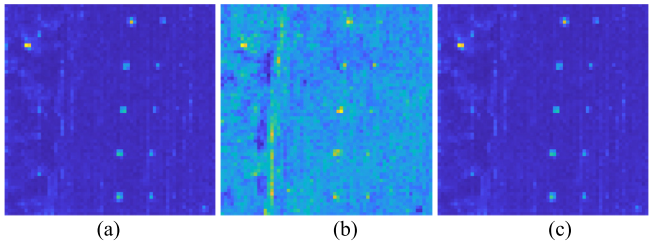


Fig. 5. AD maps of 19 R panel pixels in HYDICE data by (a) DS-AD, (b) K-AD, and (c) R-AD.

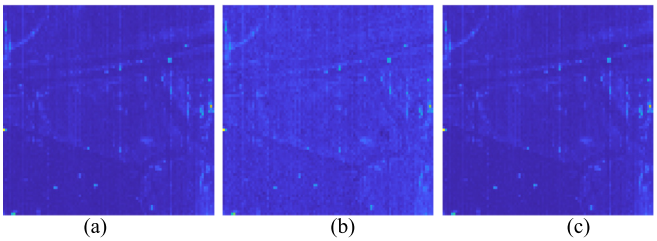


Fig. 6. AD maps of airplanes for HYDICE Urban scene by (a) DS-AD, (b) K-AD, and (c) R-AD.

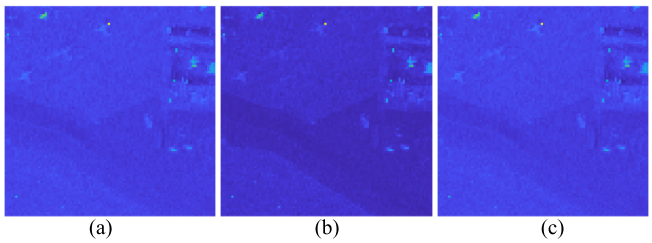


Fig. 7. AD maps of airplanes for San Diego airport by (a) DS-AD, (b) K-AD, and (c) R-AD.

the experiments. An area of 100 × 100 pixels at the upper left corner of the scene was selected as the test image shown in Fig. 4(b) along with its ground-truth map shown in Fig. 4(c). It is an urban scene in which the main BKG materials are roof, shadow, and grass. There are three airplanes in the image, which consist of 58 pixels and account for 0.33% of the test image as shown in Fig. 4(b). The spectral signatures of these planes are different from those of the main BKG as shown in Fig. 4(c), and these pixels were therefore selected as targets to be detected. This scene is less interesting than the HIDCE 15-panel data in Fig. 2 because only ground truth is available is three airplanes which are large targets.

X. Experiments and Discussion

A. AD Experiments

In the following AD experiments, we are not supposed to know the desired target signature **t**. The three anomaly detectors, DS-AD, K-AD, and R-AD were implemented for AD. Figs. 5–7 show their AD maps of HYDICE panel scene, HYDICE urban scene, and San Diego airport scene, respectively.

Apparently, it is difficult to evaluate these three anomaly detectors by visual inspection from Figs. 5–7. In this case, the detection measures derived in Section VIII would be very useful for quantitative evaluation. Table III tabulates calculated the eight 3-D ROC curve-derived detection measures,

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TABLE III
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR ROWS DETECTION RESULTS FROM FIGS. 6 AND 7

	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
HYDICE panel	DS-AD	0.9898	0.3574	**0.0430**	1.3472	**0.9466**	0.3145	**8.3205**	1.3042
	K-AD	0.9040	0.6029	0.3430	1.5069	0.5608	0.2600	1.7580	1.1639
	R-AD	**0.9900**	**0.3584**	0.0435	1.3484	0.9462	0.3149	8.2341	**1.3049**
HUDICE Urban	DS-AD	**0.9872**	0.2575	0.0361	1.2448	**0.9511**	0.2215	**7.1406**	1.2087
	K-AD	0.9862	**0.3126**	0.0772	**1.2988**	0.9089	**0.2354**	4.0493	**1.2216**
	R-AD	**0.9872**	0.2549	**0.0359**	1.2421	**0.9511**	0.2190	7.0965	1.2062
San Diego airport	DS-AD	**0.8334**	0.0910	**0.0453**	**0.9244**	**0.7861**	0.0457	**2.0101**	**0.8791**
	K-AD	0.8137	0.1546	0.0992	0.9683	0.7123	**0.0555**	1.5592	0.8692
	R-AD	0.8291	**0.0940**	0.0501	0.9231	0.7770	0.0439	1.8778	0.8730

$AUC_{(D,F)}$, $AUC_{(D,\tau)}$, $AUC_{(F,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{ANPR}, and AUC_{ODP} discussed in Section VIII based on Figs. 5–7 where the best results are boldfaced. For HYDICE data, R-AD performed slightly better than DS-AD and K-AD in terms of TD, while DS-AD performed slightly better than R-AD in in terms of BS. As for HYDICE urban scene, K-AD performed slightly better than the other two in terms of $AUC_{(D,\tau)}$ AUC_{TD}, and AUC_{TDBS}, while DS-AD and R-AD did better in $AUC_{(D,F)}$, AUC_{BS}, and AUC_{SNPR}. Regarding the San Diego scene DS-AD generally performed little better than R-AD and K-AD. By all things considered, K-AD generally did not perform as well as DS-AD and R-AD. The reason is that when K-AD is implemented, the covariance matrix $\mathbf{K}$ removes the mean $\boldsymbol{\mu}_{\mathbf{b}}$ but the data sample $\mathbf{r}$ does not. In this case, K-AD did not perform as well as DS-AD because $\mathbf{K}$ measures the variance/covariance with the mean $\mu_{\mathbf{b}}$ removed but the data samples $\mathbf{r}$ does not. On the other hand, K-AD did not perform well compared to R-AD because the $\mathbf{r}$ does not remove the mean $\boldsymbol{\mu}_{\mathbf{b}}$ in K-AD but its covariance matrix used in K-AD does. So, in either case, K-AD cannot compete against DS-AD or R-AD. More detailed studies on these three anomaly detectors will be discussed in Section X-B.

B. Dual HTD Experiments

This section is particularly included to conduct a comparative study between HAD and HTD as dual theories. Since the HYDICE urban scene lacks the ground truth about targets of interest, only HYDICE panel scene and San Diego airport scenes were used for experiments where the desired target signature $\mathbf{t}$ for these hyperspectral target detectors was obtained from the spectra in Fig. 2(c) for the HYDICE panel scene and San Diego airport scene by averaging the spectral signatures of the three airplanes, respectively.

In the SNR-based HAD in Section VII, we have operated data samples and target signatures for HAD in three different data whitened spaces, DS-space, K-Space, and R-space. A dual relationship between SNR-based hyperspectral target detectors, NAMD, NLRT, and CEM and their dual counterparts, the normalized versions of DS-AD, K-AD, and R-AD in HAD is described in Fig. 1. Specifically, by setting $\kappa = [(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})]^{-1}$ in DS-AD in (39) and GLRT in (55) we can obtain NAMD in (57). Similarly, by setting $\kappa = (\mathbf{t}^T\mathbf{R}^{-1}\mathbf{t})^{-1}$ in (68) we can also obtain CEM in (72).

It is worth noting that [10] followed the same treatment in Section VII by operating data samples in three different data whitened space, DS-space, K-Space, and R-space to also derive three SA-based target detectors in correspondence to DS-AD, K-AD, and R-AD, respectively, given by the following derivations.

1) SA in DS-Space:

$$\begin{aligned}\delta^{\mathrm{NMF}}(\mathbf{r}) &= \frac{\langle \hat{\mathbf{t}}, \hat{\mathbf{r}} \rangle}{||\hat{\mathbf{t}}||||\hat{\mathbf{r}}||} = \cos\hat{\theta} \\ &= \frac{(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})}{\left((\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{t}-\boldsymbol{\mu}_{\mathbf{b}})\right)^{1/2}\left((\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})\right)^{1/2}} \\ \Rightarrow \delta^{\mathrm{DS\text{-}SA}^2}(\mathbf{r}) &= \cos^2\hat{\theta}. \end{aligned} \tag{82}$$

2) SA in K-Space:

$$\begin{aligned}\cos\widehat{\theta} &= \frac{\langle \widehat{\mathbf{t}}, \widehat{\mathbf{r}} \rangle}{||\widehat{\mathbf{t}}||||\widehat{\mathbf{r}}||} = \frac{\mathbf{t}^T\mathbf{K}^{-1}\mathbf{r}}{\left(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}\right)^{1/2}\left(\mathbf{r}^T\mathbf{K}^{-1}\mathbf{r}\right)^{1/2}} \\ &\equiv \delta^{\mathrm{K\text{-}SA}}(\mathbf{r}) = \delta^{\mathrm{NMF}}(\mathbf{r}) \\ &\Rightarrow \delta^{\mathrm{ACE}}(\mathbf{r}) \equiv \delta^{\mathrm{K\text{-}SA}^2}(\mathbf{r}) = \frac{\left(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{r}\right)^2}{\left(\mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}\right)\left(\mathbf{r}^T\mathbf{K}^{-1}\mathbf{r}\right)} = \cos^2\widehat{\theta}. \end{aligned} \tag{83}$$

3) SA-R-Space:

$$\begin{aligned}\cos\bar{\theta} &= \frac{\langle \bar{\mathbf{t}}, \bar{\mathbf{r}} \rangle}{||\bar{\mathbf{t}}||||\bar{\mathbf{r}}||} = \frac{\mathbf{t}^T\mathbf{R}^{-1}\mathbf{r}}{\left(\mathbf{t}^T\mathbf{R}^{-1}\mathbf{t}\right)^{1/2}\left(\mathbf{r}^T\mathbf{R}^{-1}\mathbf{r}\right)^{1/2}} \\ &\Rightarrow \delta^{\mathrm{R\text{-}SA}^2}(\mathbf{r}) = \cos^2\bar{\theta} = \frac{\left(\mathbf{t}^T\mathbf{R}^{-1}\mathbf{r}\right)^2}{\left(\mathbf{t}^T\mathbf{R}^{-1}\mathbf{t}\right)\left(\mathbf{r}^T\mathbf{R}^{-1}\mathbf{r}\right)}. \end{aligned} \tag{84}$$

Interestingly, the two commonly used detectors, NMF [69] and the well-known ACE [18], [21] were actually obtained by operating data samples in the K-space as shown in (83). These three types of SA-based detectors, $\delta^{\mathrm{ACE}}(\mathbf{r}) \equiv \delta^{\mathrm{K\text{-}SA}^2}(\mathbf{r})$ in (82) operating in the DS-space, $\delta^{\mathrm{K\text{-}SA}}(\mathbf{r}) = \delta^{\mathrm{NMF}}(\mathbf{r})$ in (83) operating in the K-space and $\delta^{\mathrm{R\text{-}SA}^2}(\mathbf{r})$ in (84) operating the R-space in [10] have their corresponding counterparts in HAD, which are $\delta_{\kappa}^{\mathrm{DS\text{-}AD}}(\mathbf{r})$ in (39), $\delta_{\kappa}^{\mathrm{K\text{-}AD}}(\mathbf{r})$ in (43) and $\delta_{\kappa}^{\mathrm{R\text{-}AD}}(\mathbf{r})$ in (48), respectively.

Figs. 8(a)–(c) and 9(a)–(c) show the detection maps of 19 panel pixels in five rows and three airplanes by NAMD, NLRT, CEM, DS-SA^2, ACE, and R-SA^2, respectively.

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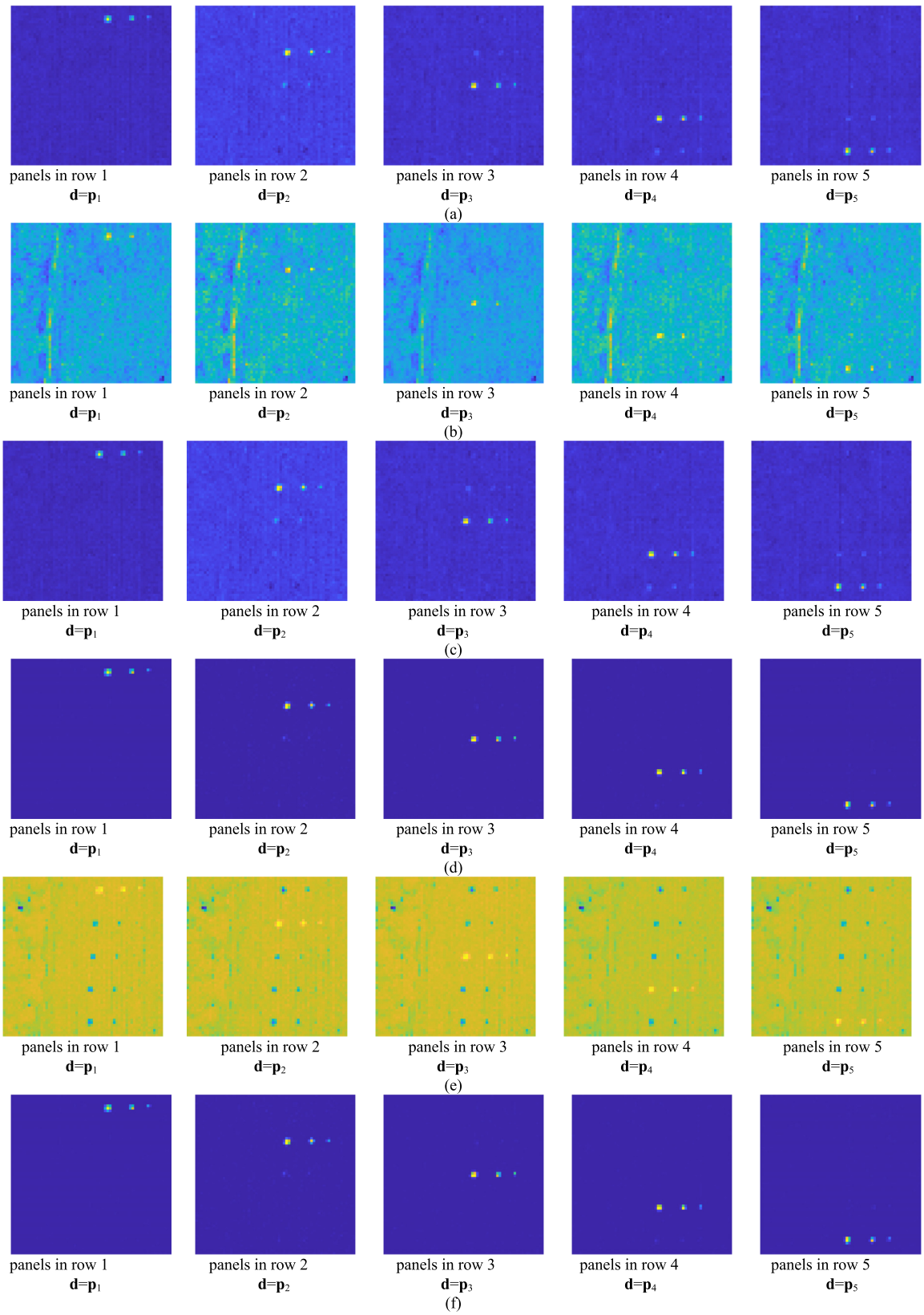


Fig. 8. Detection maps of 19-panel pixels in five rows by NAMD, NLRT, CEM, DS-SA^2, ACE, and R-SA^2.(a) Detection maps of NAMD. (b) Detection maps of NLRT. (c) Detection maps of CEM. (d) Detection maps of DS-SA^2. (e) Detection maps of ACE. (f) Detection maps of R-SA^2.

Tables IV and V also calculated their corresponding eight detection measures, $\text{AUC}_{(\text{D,F})}$, $\text{AUC}_{(\text{D},\tau)}$ $\text{AUC}_{(\text{F},\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, AUC_{SNPR}, and AUC_{ODP}.

According to Tables III–V and also Figs. 5–9, NAMD and CEM performed very closely and competitively and both outperformed NLRT. These results also had their corresponding

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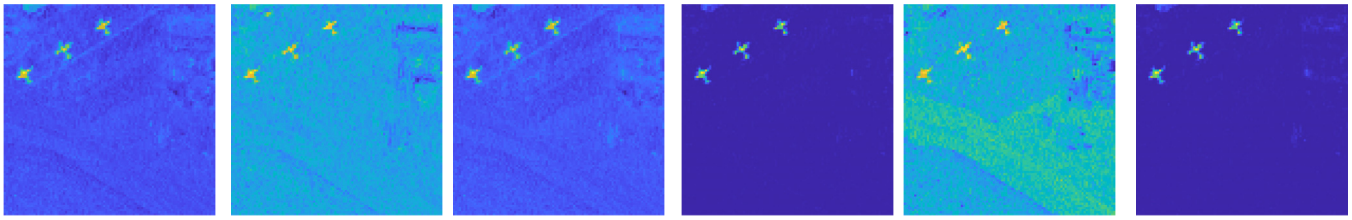


Fig. 9. Detection maps of San Diego airport data by (a) NAMD, (b) NLRT, (c) CEM, (d) DS-SA^2, (e) ACE, and (f) R-SA^2.

TABLE IV

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS OBTAINED FOR HYDICE DATA FROM FIG. 2

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
p_1	NAMD	**0.9993**	0.5947	0.0085	1.5940	0.9904	0.5863	70.2083	1.5855
	NLRT	0.9950	**0.7750**	0.3791	**1.7701**	0.6135	0.3960	2.0447	1.3910
	CEM	**0.9993**	0.5953	**0.0084**	1.5946	**0.9905**	**0.5869**	**70.8589**	**1.5862**
	DS-SA^2	**0.9994**	0.7416	0.0019	1.7411	0.9972	0.7397	385.6793	1.7392
	ACE	0.9992	0.9451	0.7554	1.9443	0.2434	0.1898	1.2512	1.1890
	R-SA^2	**0.9994**	0.7426	0.0019	1.7420	0.9972	**0.7407**	388.7317	**1.7401**
p_2	NAMD	**0.9998**	0.8310	**0.0169**	1.8309	**0.9828**	0.8141	49.0473	1.8139
	NLRT	0.9950	0.7818	0.4320	1.7768	0.5613	0.3498	1.8098	1.3448
	CEM	**0.9998**	**0.8325**	**0.0169**	**1.8323**	**0.9828**	**0.8156**	**49.2400**	**1.8154**
	DS-SA^2	0.9998	0.8135	0.0027	1.8133	0.9970	0.8108	299.2357	1.8106
	ACE	**0.9999**	0.9740	0.7503	1.9739	0.2496	0.2237	1.2982	1.2236
	R-SA^2	0.9998	0.8138	0.0027	1.8137	0.9971	0.8111	300.8682	1.8110
p_3	NAMD	**0.9997**	0.7638	**0.0108**	1.7635	**0.9888**	0.7530	70.4848	1.7527
	NLRT	0.9966	**0.8045**	0.3568	**1.8011**	0.6387	0.4477	2.2547	1.4443
	CEM	**0.9997**	0.7658	**0.0108**	1.7655	**0.9888**	**0.7550**	**70.8975**	**1.7547**
	DS-SA^2	0.9998	0.8539	0.0019	1.8537	0.9979	0.8520	448.7642	1.8518
	ACE	**0.9999**	0.9912	0.7578	1.9912	0.2421	0.2334	1.3080	1.2333
	R-SA^2	0.9998	0.8554	0.0019	1.8553	0.9979	**0.8535**	450.5786	**1.8534**
p_4	NAMD	**0.9998**	0.8149	0.0105	1.8147	0.9892	0.8044	77.3337	1.8042
	NLRT	0.9729	**0.8526**	0.4520	**1.8255**	0.5119	0.4006	1.8862	1.3735
	CEM	**0.9998**	0.8149	**0.0104**	1.8147	**0.9894**	**0.8045**	**78.5316**	**1.8044**
	DS-SA^2	**0.9998**	0.8444	0.0018	1.8442	0.9980	0.8426	478.3755	1.8425
	ACE	**0.9998**	0.9579	0.7350	1.9577	0.2647	0.2228	1.3032	1.2227
	R-SA^2	**0.9998**	0.8448	0.0017	1.8446	0.9980	**0.8431**	484.2071	1.8429
p_5	NAMD	**0.9998**	0.7372	0.0104	1.7369	0.9893	0.7268	71.0737	1.7266
	NLRT	0.9974	**0.8635**	0.4157	**1.8610**	0.5808	0.4478	2.0772	1.4452
	CEM	**0.9998**	0.7376	**0.0103**	1.7373	**0.9894**	**0.7273**	**71.4553**	**1.7270**
	DS-SA^2	0.9998	0.7742	0.0018	1.7739	0.9979	0.7724	434.8178	1.7722
	ACE	0.9998	0.9528	0.7398	1.9526	0.2600	0.2130	1.2879	1.2128
	R-SA^2	0.9998	0.7747	0.0018	1.7744	0.9979	**0.7729**	438.8971	**1.7727**

TABLE V

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS OBTAINED FOR SAN DIEGO AIRPORT FROM FIG. 5

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}
NAMD	0.9766	0.4768	**0.0295**	1.4533	0.9468	0.4473	**16.1793**	1.4239
NLRT	0.9135	**0.6576**	0.3775	**1.5711**	0.5349	0.2801	1.7418	1.1936
CEM	**0.9901**	0.4862	0.0301	1.4763	0.9599	**0.4561**	16.1598	**1.4462**
ACE	0.8958	0.6961	0.4344	1.5920	0.4602	0.2617	1.6025	1.1576
DS-SA^2	0.9772	0.4032	0.0038	1.3803	0.9731	0.3994	106.8681	1.3766
R-SA^2	**0.9911**	0.4087	0.0041	1.3998	0.9868	0.4046	98.7384	1.3956

dual results in Table II and Figs. 5–7. More specifically, as for HAD, DS-AD (i.e., RXD), and R-AD are the best anomaly detectors. Correspondingly, NAMD and CEM were also among the best hyperspectral target detectors which were also confirmed and verified in [10]. Most interestingly, ACE was claimed to the best detector in [21] turned out to be

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the worst detector. However, according to Tables IV and V, its values of $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, and $\mathrm{AUC}_{(\mathrm{F},\tau)}$ were very high but its other five values of $\mathrm{AUC}_{(\mathrm{F},\tau)}$, $\mathrm{AUC}_{\mathrm{TD}}$, $\mathrm{AUC}_{\mathrm{BS}}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, $\mathrm{AUC}_{\mathrm{SNPR}}$, and $\mathrm{AUC}_{\mathrm{ODP}}$ were, indeed, very low. In particular, $\mathrm{AUC}_{(\mathrm{F},\tau)}$ was supposed to be low to reflect its BKG suppressibility. But ACE had very high $\mathrm{AUC}_{(\mathrm{F},\tau)}$ values, as shown in Figs. 8 and 9. This implied that ACE was, indeed, not an effective detector. Unfortunately, if only $\mathrm{AUC}_{(\mathrm{D,F})}$ values are used without $\mathrm{AUC}_{(\mathrm{F},\tau)}$ values to evaluate a detector, it is very likely to lead to an incorrect conclusion as reported by many works in the literature.

Similar phenomena to Tables IV and V were also observed from Table III where K-AD corresponding to the LRT-based NLRT and SA-based K-SA[2] in HTD also produced very high values of $\mathrm{AUC}_{(\mathrm{D,F})}$, $\mathrm{AUC}_{(\mathrm{D},\tau)}$, and $\mathrm{AUC}_{\mathrm{TD}}$, but very poor values of $\mathrm{AUC}_{\mathrm{BS}}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, $\mathrm{AUC}_{\mathrm{SNPR}}$, and $\mathrm{AUC}_{\mathrm{ODP}}$. So, once again, if only $\mathrm{AUC}_{(\mathrm{D,F})}$ is used for performance evaluation, the conclusion is also misleading.

As a conclusion, what $\delta^{\text{K-AD}}(\mathbf{r})$ is to HAD is exactly the same as what $\delta^{\text{K-SA}^2}(\mathbf{r}) = \delta^{\text{ACE}}(\mathbf{r})/\delta^{\text{K-SA}}(\mathbf{r}) = \delta^{\text{NMF}}(\mathbf{r})$ is to HTD. If $\delta^{\text{K-AD}}(\mathbf{r})$ performed poorly for HAD, then so did the SA-based $\delta^{\text{ACE}}(\mathbf{r})$ and LRT-based $\delta^{\text{NLRT}}(\mathbf{r})$ for HTD. On the other hand, both $\delta^{\text{DS-AD}}(\mathbf{r}) = \delta^{\text{RXD}}(\mathbf{r})$ and $\delta^{\text{R-AD}}(\mathbf{r})$ performed well and comparably for HAD as shown in Table II and Figs. 5–7, so did $\delta^{\text{NAMD}}(\mathbf{r})$ and $\delta^{\text{CEM}}(\mathbf{r})$, as shown in Tables IV and V and Figs. 8 and 9. These experiments explained why many anomaly detectors have been derived from either $\delta^{\text{RXD}}(\mathbf{r})$ or $\delta^{\text{R-AD}}(\mathbf{r})$ and many target detectors are also derived from $\delta^{\text{CEM}}(\mathbf{r})$ such as hierarchical CEM (H-CEM) [70] and ensemble-based cascaded CEM (E-CEM) [71]. According to our experiments for HAD, such extensions should be also work as well as $\delta^{\text{CEM}}(\mathbf{r})$-extended target detectors do for HTD. Interestingly, there was no corresponding work reported in the literature on extending $\delta^{\text{NAMD}}(\mathbf{r})$ (i.e., normalized Mahalanobis distance) in HTD.

XI. Conclusion

Hypothesis testing theory is generally used to develop LRT-based detectors for HTD. However, such LRT-derived target detectors require the targets under the alternative hypothesis to be known. To address this issue, GLRT was developed to derive the well-known RXD anomaly detector by making two assumptions on: 1) the targets to be known with their signatures, that is, target shapes, but with unknown target signal intensities (i.e., abundance fractions) which need to be estimated and 2) availability of a reference dataset from which target signal intensities can be estimated. In addition, the probability distribution under each hypothesis must also be provided such as Gaussian distributions. To relax these two assumptions, an SNR-based theory was developed for all LRT-based detectors in [10]. Interestingly, the same theory can be also applied to HAD as shown in Sections V and VI. Furthermore, since both assumptions are generally not valid for HAD, LRT-derived detectors cannot be directly applied to HAD. This article further develops a dual theory of HTD for HAD, which converts HTD to HAD, specifically, making LRT-based target detectors anomaly detectors. In particular, RXD can be shown to be a special case of SNR-derived anomaly detectors. The most significant contributions are summarized as follows.

1) This article develops HTD and HAD are dual theories, which have never been reported in the literature.
2) Compared to RXD derived from the very complicated hypothesis testing theory-based GLRT, it offers simple and intuitive approaches to dual theories of HTD and HAD with detailed theoretical derivations for RXD to establish their close relationships and connections.
3) This article takes advantage of SNR to develop an SNR-based theory for HAD and also rederives LRT-based detectors including GLRT as its special case.
4) This article reinterprets $\delta^{\text{GLRT}}(\mathbf{r})$ as a function of two dual variables, data sample $\mathbf{r}$ and target signature $\mathbf{t}$, denoted by $\delta^{\text{GLRT}}_{\mathbf{t}}(\mathbf{r})$. For HTD, $\delta^{\text{GLRT}}_{\mathbf{t}}(\mathbf{r})$ is a linear MF of $\mathbf{r}$ using the specifically known target signature $\mathbf{t}$ as its matching signal in which case, $\max_{\mathbf{r}} \text{SNR}\{\delta^{\text{GLRT}}_{\mathbf{t}}(\mathbf{r})\} = \mathbf{t}^T\mathbf{K}^{-1}\mathbf{t}$ with optimal solution $\mathbf{r}$ is obtained by its matching signal $\mathbf{t}$. On the other hand, since $\mathbf{t}$ is unknown and $\mathbf{r}$ is a given data sample $\mathbf{r}$ in HAD, the best and optimal solution to $\delta^{\text{GLRT}}_{\mathbf{t}}(\mathbf{r})$ is, indeed, a K-/R-correlator or correlated filter using $\mathbf{r}$ as its matching signal, $\mathbf{t}$ which results in a quadratic filter of $\mathbf{r}$. In this case, depending upon on the space in which $\mathbf{r}$ is operated, there are three scenarios.
   a) $\max_{\mathbf{t}} \text{SNR}\{\delta^{\text{DS-AD}}_{\mathbf{t}}(\mathbf{r})\} = (\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})^T\mathbf{K}^{-1}(\mathbf{r}-\boldsymbol{\mu}_{\mathbf{b}})$ specified by (31) in the DS-space, which turns out to be RXD in [12] and also known as Mahalanobis distance;
   b) $\max_{\mathbf{t}} \text{SNR}\{\delta^{\text{K-AD}}_{\mathbf{t}}(\mathbf{r})\} = \mathbf{r}^T\mathbf{K}^{-1}\mathbf{r}$ in the K-space;
   c) $\max_{\mathbf{t}} \text{SNR}\{\delta^{\text{R-AD}}_{\mathbf{t}}(\mathbf{r})\} = \mathbf{r}^T\mathbf{R}^{-1}\mathbf{r}$ in the R-space which becomes R-AD in (75) developed in [67].
5) It should be noted that the optimal target detector for HTD uses the known target signal $\mathbf{t}$ as the matched signal to derive a linear MF. As for HAD, since there is no known target signal $\mathbf{t}$ a hyperspectral anomaly detector must use itself as the matched signal to derive a quadratic detector. This makes sense because self-correlated filter performs better than cross-correlated filter. This is how an anomaly detector turns out to be a quadratic detector.
6) Finally, in spite of the fact that many works on HTD and HAD were reported in hyperspectral imaging, it is our belief that there is still a need for a fundamental treatment on these topics. This article can be considered as a companion work of [10] where both together along with [23] lay out the foundation of HTD and HAD theories. Those who are working in these areas should find benefits from these three papers to fill in many missing puzzles in their BKG.

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Chein-I. Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York at Stony Brook, Stony Brook, NY, USA, in 1977, the M.S. and M.S.E.E. degrees from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, in 1982, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He has been with the University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, since 1987, where he is a Professor with the Department of Computer Science and Electrical Engineering. He is holding a Chang Jiang Scholar Chair Professorship and has been the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, since 2016. In addition, he also has been a Chair Professor of National Chiao Tung University, Hsinchu, since 2019. He has authored over four books, including *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing*: *Endmember Finding and Anomaly Detection* (Springer, 2016), and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017). In addition, he also edited two books, *Recent Advances in Hyperspectral Signal and Image Processing*, (John Wiley & Sons, 2006) and *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007) and co-edited with A. Plaza, *High Performance Computing in Remote Sensing* (CRC Press, 2007). His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE. He received the National Research Council Senior Research Associateship Award from 2002 to 2003 sponsored by the U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, MD, USA. He serves as an Associate Editor for *Remote Sensing* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

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